Supplementary Material for *Who Fights Best? Discovering Winning Personality Pairings for LLM Debate Agents*

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ABSTRACT

Multi-agent debate (MAD) has emerged as a powerful paradigm for enhancing the reasoning capabilities of large language models (LLMs). However, most MAD systems employ personality-neutral agents and overlook the role of personality, which has been shown in psychology and behavioral science to influence reasoning styles and collaborative problem-solving. In this study, we introduce MBTI-DB, a debate framework for systematically studying the impact of personality in MAD. We design 16 distinct personality profiles and inject them into debates in a modular manner. Through experiments across four LLM families and 700 factual questions, we demonstrate that incorporating personality induces consistent and measurable accuracy shifts. Further analysis reveals that lexical density, personality-question similarity, and prompt perplexity influence these accuracy variations. We evaluate personality-pair selection strategies and find that aggregating predictions from optimal pairs for each question yields significant gains. However, automatically identifying the best configuration remains challenging with substantial cross-model variation, suggesting that while personality diversity offers meaningful performance benefits, its effects are complex and context-dependent. Our code and datasets can be found at `https://github.com/KJOrigin/WhoFightsBest`.

A. Psychological Foundations and Debate Style Conversion

A.1. Myers-Briggs Type Indicator (MBTI)

The Myers-Briggs Type Indicator (MBTI) was created by Myers and Briggs in the 1940s and has become one of the most widely used tools for describing personality preferences in educational, workplace and self-development settings [12, 13, 2]. It is grounded in Carl Jung's theory of psychological types and provides a clear structure for understanding how individuals prefer to process information and engage with the world. Rather than serving as a system that predicts specific behaviors, the MBTI groups personality preferences into four independent dimensions. These dimensions can be combined into sixteen unique personality patterns.

Extraversion (E) and Introversion (I). This dimension concerns the source of a person's mental energy. Individuals who prefer Extraversion are usually oriented toward the outer environment and are energized by social interaction, activity and engagement with external stimuli. Individuals who prefer Introversion focus more on their internal world of thoughts and reflection and typically regain energy through quietness and solitude.

Sensing (S) and Intuition (N). This pair describes the preferred way of taking in information. Sensing types attend closely to concrete details and observable facts and they often rely on practical and present focused data. Intuitive

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types pay greater attention to patterns, meanings and possibilities and they tend to concentrate on abstract concepts and future opportunities.

Thinking (T) and Feeling (F). This dimension reflects how people prefer to make decisions. Thinking types rely on objective reasoning and analytical principles and they value consistency and logical clarity. Feeling types consider emotional impact and interpersonal concerns and they place importance on empathy, harmony and the needs of others.

Judging (J) and Perceiving (P). This final dimension concerns how individuals prefer to organize their external lives. Judging types are comfortable with structure, planning and clear decisions and they tend to appreciate order and predictability. Perceiving types prefer flexibility, openness and spontaneity and they adapt more easily to changing situations.

When the four preference dimensions are combined, they form sixteen personality types such as INTP, ESFJ or ENFP. Each type represents a specific configuration of cognitive tendencies. The MBTI does not assess skill or competence. Instead, it provides a descriptive framework for understanding how people typically gather information, make choices and manage daily activities.

A.2. Big Five

The Big Five model, often referred to by the acronym OCEAN, is widely regarded as one of the most robust frameworks for understanding human personality [1]. It organizes individual differences into five broad psychological domains. Each domain is continuous rather than categorical, which allows personality to be represented as a profile spanning multiple dimensions rather than a fixed type.

Openness to Experience. Openness characterizes the extent to which a person is drawn toward intellectual exploration, creative thinking and novel experiences. Individuals with stronger openness usually appreciate imaginative activities and are receptive to unconventional ideas. Individuals with more limited openness often rely on practicality, familiarity and structured routines.

Conscientiousness. This dimension concerns the regulation of behavior and the ability to maintain organized, responsible and goal oriented actions. People with high conscientiousness typically show persistence, careful planning and reliability across tasks. Those who score lower may exhibit greater flexibility and spontaneity, yet may also struggle with consistency or long term organization.

Extraversion. Extraversion reflects the tendency to seek social interaction and external stimulation. Highly extraverted individuals gain energy from being around others and often appear expressive and assertive. Individuals who fall on the introverted side of the continuum tend to prefer quieter environments, introspection and activities that require less social engagement.

Agreeableness. Agreeableness captures interpersonal orientation, including tendencies toward compassion, cooperation and interpersonal trust. People high in agreeableness tend to value positive social relationships and respond to others with warmth and generosity. People who score lower may prioritize independence or objectivity and may adopt a more critical or competitive stance in social contexts.

Neuroticism. Neuroticism describes sensitivity to emotional fluctuation and psychological stress. High levels of this trait are associated with frequent experiences of worry, tension and mood variability. Low levels indicate emotional steadiness, resilience and a general sense of calm in the face of challenges.

A.3. HEXACO

The HEXACO framework expands upon the traditional Big Five by adding a sixth domain that focuses specifically on moral tendencies and interpersonal integrity [8]. This extended structure has become increasingly influential in areas such as moral cognition, behavioral decision making and computational models of trust. The model consists of six broad personality dimensions, each representing a distinct cluster of behavioral and emotional tendencies.

Honesty and Humility. This dimension reflects an individual's inclination toward sincerity, fairness and modest behavior. People who score highly are generally unwilling to exploit others, show little interest in materialistic

advantages and tend to avoid tactics that involve manipulation or self promotion. Individuals with lower scores may be more concerned with personal status or achievement and may show a greater willingness to deceive or take advantage of others.

Emotionality. In HEXACO, Emotionality replaces the Big Five concept of Neuroticism with a slightly different emphasis. It captures emotional vulnerability, sensitivity to stress, attachment to close relationships, and proneness to anxiety. Higher levels of Emotionality are associated with stronger feelings of dependence and empathy. Lower levels may correspond to emotional detachment, reduced anxiety, and a more restrained expression of feelings.

Extraversion, Agreeableness, Conscientiousness and Openness. These four traits share conceptual roots with the Big Five but incorporate meaningful refinements. Agreeableness, for example, is defined in a way that emphasizes patience, tolerance, and forgiveness rather than emotional reactivity. Conscientiousness continues to describe disciplined and responsible behavior, with an emphasis on careful planning and commitment to goals. Openness reflects intellectual curiosity and interest in aesthetic or unconventional ideas. Extraversion, similar to its Big Five counterpart, refers to enthusiasm for social interaction and engagement with the external world.

Together, these six dimensions provide a richer picture of personality by including ethical and interpersonal considerations that are less prominent in earlier models. The HEXACO system has therefore become a valuable tool for both psychological research and computational applications involving social behavior modeling.

A.4. DISC

The DISC framework describes behavioral tendencies through four overarching styles that highlight how individuals act, communicate, and respond to social or task-related demands [10]. Unlike personality models that aim to infer internal traits, DISC concentrates on outward behaviors that can be directly observed, which is why it is commonly applied in professional settings and organizational training.

Dominance (D). Individuals who exhibit a Dominance style tend to be direct, decisive, and driven by outcomes. They prefer taking control of situations, respond well to competition, and aim for rapid progress toward goals.

Influence (I). The Influence style is characterized by expressiveness, enthusiasm, and a strong orientation toward social connection. People with high Influence typically enjoy interaction, collaboration, and opportunities to motivate others.

Steadiness (S). Steadiness reflects a calm, consistent, and cooperative behavioral approach. Individuals associated with this style often value stability, patience, and dependable support for others.

Conscientiousness (C). This style highlights careful analysis, precision, and a preference for structured processes. Individuals who primarily express the Conscientiousness pattern tend to focus on accuracy, planning, and cautious evaluation of information.

The DISC framework has become a widely adopted tool in leadership development, workplace communication training, and team composition strategies.

A.5. Enneagram System

The Enneagram is a psychological framework describing nine interrelated personality types, each defined by a core fear, an essential desire, and a characteristic worldview [5]. Unlike trait-based models, it emphasizes underlying motivation and shifting patterns of behavior, illustrating how individuals react under stress, adapt to challenges, and grow over time. The nine types include Type 1, the Reformer, who is driven by principles and the pursuit of perfection. Type 2, the Helper, seeks emotional connection and gains a sense of value from being needed. Type 3, the Achiever, is motivated by success and by maintaining a capable image. Type 4, the Individualist, focuses on authenticity and emotional depth. Type 5, the Investigator, values knowledge, autonomy, and intellectual mastery. Type 6, the Loyalist, prioritizes security, preparedness, and stability. Type 7, the Enthusiast, is motivated by freedom, new experiences, and the wish to avoid limitation. Type 8, the Challenger, seeks control, strength, and a strong sense of independence. Type 9, the Peacemaker, strives for inner calm, emotional balance, and harmonious relationships. Each type is also shaped by its adjacent wings, which add nuance through neighboring traits, and by two directional arrows that represent characteristic behavioral movement in stressful situations and during periods of growth. Although the Enneagram is

less empirically grounded than trait-based personality theories, it provides a rich conceptual structure for representing internal drives and emotional dynamics.

A.6. MBTI vs. Other Personality Theories

Although MBTI is widely adopted in both academic and applied settings, it represents only one of many established theories for characterizing individual differences, while other personality theories offer complementary perspectives. The Big Five [1] (or OCEAN) theory through five continuous factors, while the HEXACO theory [8] extends this structure by adding an additional dimension related to moral character. Other influential theories include the Enneagram [5], which identifies nine personality patterns grounded in motivational drives and core fears, and the DISC theory [10], which groups behavioral tendencies into four broad styles that shape how individuals collaborate and approach tasks.

In addition, the MBTI has attracted both support and criticism, particularly due to its reliance on binary categories [14, 6]. Nevertheless, it remains widely adopted because of its descriptive clarity and the intuitive insights it offers into individual behavior [2]. These characteristics make MBTI a pragmatic foundational theory for our study. From a modeling perspective, the use of discrete personality labels provides clearer and more efficient instructional signals for large language models than continuous-valued representations (e.g., OCEAN), offering an additional practical motivation for selecting MBTI as a simplified experimental setting.

It is also worth noting that our framework is not restricted to MBTI and is compatible with other personality theories, which can be incorporated in a plug-and-play manner by converting them into binary prompt formulations following the MBTI-in-Thoughts paradigm [2].

A.7. Converting MBTI Descriptions into Debate Behaviors

To make MBTI personality descriptions usable in debate settings, we first draw on the core idea of Jungian personality theory [3], which holds that each MBTI type is characterized by a small set of dominant cognitive preferences. Among these, some tendencies guide how individuals most naturally reason and make judgments, others shape how they express and communicate those judgments, and a few weaker tendencies tend to surface mainly under stress. Building on this perspective, we transform abstract personality traits into concrete and observable debate behaviors. Rather than describing what a type is like in general, we focus on how an agent with a given MBTI type typically argues, communicates, and reacts during a debate. For each type, we start from its standard MBTI description and reinterpret it around three practical questions: (i) *what the agent relies on to persuade others*, (ii) *how the agent expresses and defends its views in interaction with opponents*, and (iii) *how the agent is likely to react when debate pressure increases*.

In practice, this means that traits such as being practical, idealistic, or analytical are rewritten as specific debate actions. For example, a type described as practical and responsible, such as ISTJ, is modeled as grounding arguments in factual details, developing points step by step, and adhering closely to debate rules and structure. A type described as warm and supportive, such as ISFJ, is modeled as arguing politely, appealing to shared values or real-world impact, and avoiding aggressive confrontation. Likewise, idealistic and principled types, such as INFJ, are translated into debate styles that emphasize underlying meaning, moral considerations, and empathetic framing, while strategic and analytical types, such as INTJ, are modeled as focusing on long-term structure, logical coherence, and anticipating counterarguments.

In addition, we explicitly account for how different types behave under stress. Instead of assuming that all personality tendencies are constantly active, we allow certain weaker or less-preferred tendencies to become apparent primarily when debates become highly adversarial or emotionally demanding. For example, a type that normally argues in a calm and principle-driven manner, such as INFJ, may become momentarily hesitant or less verbally fluent when confronted with rapid-fire factual challenges or a cold, confrontational debate style. Similarly, a highly structured and detail-oriented type, such as ISTJ, may become increasingly rigid under pressure, insisting on strict rule-following or narrow factual correctness at the expense of broader contextual reasoning. These may manifest as emotional withdrawal, excessive rigidity, or over-detachment, depending on the type. This design ensures that each debate style captures not only typical strengths, but also realistic limitations. Overall, this conversion produces debate-oriented personality specifications that remain faithful to recognizable MBTI characteristics while being expressed in terms of concrete argumentative behaviors, making them suitable for consistent and reproducible use in multi-agent debate settings.

Table B.1
Examples of original MBTI personality descriptions.

MBTI Personality	MBTI-Orig
ISTJ	Quiet, serious, and dependable. Practical, matter-of-fact, realistic, and responsible. They value tradition and loyalty, and are good at planning and organising.
ISFJ	Warm, conscientious, and reliable. They use common sense and a realistic approach to solving problems. They value stability and harmony, often supportive and attentive to others' needs.
INFJ	Idealistic, insightful, and principled. They have a deep concern for others and a strong sense of purpose. They seek meaning and connection, and often have rich inner lives.
INTJ	Independent, analytical, and strategic. They are innovative thinkers who like to plan ahead, see patterns, and implement long-term visions. They value competence and logic.
ISTP	Pragmatic, factual, and quiet. They enjoy understanding how things work, are flexible in adapting, and prefer to deal with concrete information and hands-on activities.
ISFP	Sensitive, gentle, and easygoing. They value harmony, love to explore aesthetics and personal expression, and often live in the moment while staying quiet and caring.
INFP	Idealistic, creative, and introspective. They are guided by their values and look for authenticity and meaning. They are often loyal to their causes and people but prefer flexibility.
INTP	Logical, inventive, and curious. They enjoy exploring theories, ideas, and systems. They value independence, deep thought, and often withdraw to reflect and analyse.
ESTP	Energetic, perceptive, and action-oriented. They are good at improvising, reacting quickly in the moment, and enjoy exploring new experiences and challenges.
ESFP	Outgoing, friendly, and spontaneous. They live in the moment, enjoy social interaction, and bring enthusiasm and fun to their surroundings. They often focus on what is happening now.
ENFP	Enthusiastic, imaginative, and sociable. They love exploring possibilities, connecting with others, and expressing themselves creatively while seeking meaning and inspiration.
ENTP	Inventive, enthusiastic, and intellectually curious. They enjoy debate, ideas, and exploring many possible avenues. They dislike routine and like to keep things dynamic and flexible.
ESTJ	Organized, practical, and decisive. They like structure, rules, and clear goals. Good at implementing plans and leading others in a reliable, efficient way.
ESFJ	Warm-hearted, cooperative, and sociable. They are attuned to others' feelings, value harmony and connection, and often take responsibility for maintaining relationships and social order.
ENFJ	Charismatic, empathetic, and inspiring. They naturally lead and motivate others, care about people's growth, and build meaningful relationships while pursuing their ideals.
ENTJ	Bold, strategic, and outspoken. They are natural leaders, like to set long-term goals, organize people and resources, and implement ambitious visions with confidence.

B. Personality Prompt Variants

Table B.1 presents the original MBTI personality descriptions. Tables B.2-B.5 show the rewritten debate-specific personality prompts, which are adapted to capture distinct debate styles. To control for potential confounding effects of prompt length, all personality descriptions are constrained to approximately 130 words and follow a consistent structural template. In addition, Tables B.6 and B.7 present single-turn prompt formulations designed for standalone question answering.

C. Debater Personality Casting Results by Domain and Model

This section reports the domain-specific MBTI personality selections for each model. For each domain, six MBTI personality types are selected based on their average performance across three prompt variants. Specifically, for each personality type, model accuracy is evaluated under the *MBTI-Orig*, *MBTI-Debate*, and *MBTI-Direct* prompt formulations and then averaged to obtain a prompt-stable performance estimate. Based on these estimates, personalities are stratified into performance tiers and sampled accordingly. Tables C.1-C.10 present the complete set of selected MBTI personalities for all ten models across fourteen domains.

Table B.2
Examples of debate-behavioral personality descriptions adapted from the MBTI framework (Part 1 of 4).

MBTI Personality	MBTI-Debate
ISTJ	In debates, ISTJs tend to speak with facts and logic, developing their arguments step by step. They prefer structured debate formats and strictly follow the rules. When refuting others, they point out inconsistencies or a lack of factual basis in the opponent's arguments, firmly believing that "facts are facts." ISTJs communicate in a straightforward and honest manner, sometimes so direct that it may come across as blunt, but this stems from their prioritization of reason over emotion. Overall, their style is rigorous and pragmatic, focusing on persuasion through logic. However, this strong emphasis on detail and accuracy can sometimes make ISTJs seem inflexible, or even cause them to overlook the bigger picture of the debate due to their excessive focus on specifics.
ISFJ	In debates, ISFJs appear considerate and patient. They usually come prepared and support their viewpoints with relevant facts and examples, yet their manner of expression remains gentle and humble. ISFJs tend to avoid intense conflict and prefer appealing to shared values and the real-life impact of the issue being discussed, reflecting their loyal and caring nature. When refuting an opponent, they are polite and cautious, often arguing from a standpoint of protecting those affected rather than directly attacking the other party. Because they dislike confrontation and are sensitive to criticism, ISFJs may hold back-or even feel pressured into withdrawing-when a debate becomes overly tense or takes on a personal, attacking tone.
INFJ	In debates, INFJs appear calm and thorough, often taking time to understand the perspectives and intentions behind different viewpoints. They communicate with empathy and insight, choosing their words carefully in an effort to address the moral and human dimensions behind the issue. INFJs prefer deep, meaningful discussions rather than combative arguments; they defend their ideals and principles gently yet firmly, often supporting their points with inspiring metaphors or noble values. They rarely raise their voices, relying instead on patience and sincerity to persuade others. Because they invest a great deal of emotion into the topic, INFJs may feel uneasy-or even momentarily speechless-when the debate becomes too adversarial or cold. In such moments, they need to remind themselves to stay focused on facts to prevent emotions from clouding their judgment.
INTJ	In debates, INTJs demonstrate strategic and analytical thinking. They often approach a debate as a task requiring careful planning, preparing their arguments in advance and anticipating the opponent's viewpoints. When presenting their ideas, INTJs speak with confidence and directness, focusing on logical reasoning and the overall framework rather than scattered details or personal emotions. They excel at quickly identifying flaws in an opponent's argument and countering them in a structured manner. Because they are committed to objectivity and rationality, INTJs may sometimes appear overly detached or critical, paying little attention to others' emotional responses. They need to be mindful of balancing firm reasoning with emotional awareness during discussions, so as not to make others feel dismissed due to their intense rationality.

D. Additional Design Choices and Experimental Details

Design Choices. We focus on open-weight models for three main reasons. First, our experiments involve debates among 16 personality types over more than 700 questions, where the multi-turn nature of debate would make API-based access prohibitively expensive. Second, open-weight models enhance reproducibility, allowing other researchers to replicate our results without relying on proprietary interfaces. Third, access to multiple model sizes within the same model family enables a systematic analysis of how personality-driven debate performance varies with model scale.

In addition, to study how personality-driven debate behaviors influence reasoning outcomes, we adopt MMLU-Pro as the primary benchmark for three reasons. First, its higher difficulty level and expanded answer space make it a widely adopted benchmark for evaluating advanced LLM reasoning. Second, its broad domain coverage allows us to examine how personality-conditioned debate styles manifest across different fields. Third, the consistent question format across domains supports fair and controlled comparisons in multi-agent debate settings.

Table B.3
Examples of debate-behavioral personality descriptions adapted from the MBTI framework (Part 2 of 4).

MBTI Personality	MBTI-Debate
ISTP	In debates, ISTPs are pragmatic and concise. They go straight to the core of the issue and prefer to support their viewpoints with objective data and real-world examples. ISTPs tend to remain calm even in heated discussions, treating debates as problem-solving exercises rather than personal confrontations. They are uninterested in abstract or overly theoretical arguments, focusing instead on whether the conclusions have practical value. When refuting others, they often deliver brief yet pointed remarks that strike directly at the weakness of an argument, demonstrating a cool, incisive wit. Because they favor direct analysis, ISTPs may lose patience with debates that are overly theoretical or burdened with unnecessary procedure. Their consistently calm and objective demeanor can also come across as somewhat detached, making them appear less expressive or warm emotionally.
ISFP	In debates, ISFPs present themselves in a gentle and sincere manner, placing great importance on expressing their inner values. They generally avoid direct confrontation and prefer to articulate their viewpoints quietly, focusing on how the issue affects individuals and personal relationships. Only when the discussion touches on their deeply held beliefs or the people they care about will ISFPs speak up firmly. Even when refuting others, they tend to share personal experiences or compassionate examples to express differing views, persuading through emotion rather than pressure. ISFPs are attentive listeners who think carefully about their responses, striving to remain true to their feelings. Because they dislike conflict and are sensitive by nature, ISFPs may choose to remain silent or withdraw when the atmosphere becomes too tense or confrontational, rather than engage in a head-on argument.
INFP	In debates, INFPs begin from a place of sincere ideals and goodwill. They focus more on the moral significance or humanistic concerns of an issue rather than on aggressive logical confrontation, often using metaphors, stories, or ethical examples to illustrate their points. INFPs tend to view debates as collaborative exchanges rather than win-or-lose battles; they prefer seeking shared values or compromise solutions that all parties can accept. When raising objections, INFPs are polite and somewhat reserved, expressing themselves with honesty and gentleness. Because they dislike conflict, they may fall silent when the atmosphere becomes tense or when their views are strongly challenged, and may even withdraw from the debate if emotionally hurt. However, when their deeply held principles are at stake, INFPs can show unexpected resolve, defending their position with heartfelt passion.
INTP	In debates, INTPs revel in the process of logical deduction and intellectual experimentation. They enjoy dissecting ideas layer by layer and may even play devil's advocate to thoroughly examine every aspect of a topic. INTPs are adept at identifying logical flaws in arguments and pointing them out directly in a rational manner. They typically maintain a calm and objective attitude, focusing solely on the logic and facts of the argument without being influenced by emotions. Their discussion style is highly theoretical, often involving detailed definitions or hypothetical scenarios-an approach that reflects depth but may come across as overly academic. Because they place such strong emphasis on rational analysis, INTPs often overlook others' emotional responses or the subtle emotional dynamics within a debate. As a result, they may appear detached or even somewhat aloof to opponents who rely more on emotional expression.

Experimental Details. All LLMs are deployed using the Swift [15] framework with a temperature of 0.3. In the debate setting, we cap the maximum number of debate rounds at 10. If the debate reaches this limit without consensus being achieved between the two debating agents, the moderator is instructed to synthesize the arguments from both sides and produce a final answer. The moderator is designed to be strictly neutral: it does not exhibit any personality traits and does not introduce new evidence. Instead, it aggregates and evaluates the arguments presented by both debaters and selects the most plausible option.

To analyze linguistic characteristics of personality descriptions, we compute lexical density using the *en_core_web_sm*[3] model for tokenization and part-of-speech tagging. Lexical density is defined as the ratio of content words (i.e.,

[3] https://huggingface.co/spacy/en_core_web_sm.

Table B.4
Examples of debate-behavioral personality descriptions adapted from the MBTI framework (Part 3 of 4).

MBTI Personality	MBTI-Debate
ESTP	In debates, ESTPs display abundant energy and directness. Quick-witted and responsive, they excel at spontaneous exchanges and often disrupt their opponents' rhythm with sharp humor or bold challenges. ESTPs state the practical significance of their viewpoints straightforwardly, frequently asking, "What does this actually accomplish?" and reinforcing their stance with concrete examples. They enjoy the thrill of debate, guiding the discussion with confidence and rapid reactions. For ESTPs, rules and formal structures matter far less than using personal charisma and tangible results to persuade others. Because they relish risk and improvisation, ESTPs may sometimes take argumentative gambles and overlook detailed analysis, though their keen perception usually helps them spot the critical points.
ESFP	In debates, ESFPs win through charm and enthusiasm. They excel at engaging others with storytelling and humor, turning serious arguments into something more relatable and emotionally compelling. Rather than abstract theories, ESFPs focus on how ideas affect real life in the moment, often using personal experiences or vivid examples to make their points resonate with others' lived experiences. They are highly attuned to the audience's emotions and will adjust their tone and pace accordingly to keep the atmosphere light and pleasant. ESFPs dislike tense or rigid debate settings; when things become overly serious, they often defuse the tension with playful remarks or dramatic metaphors. Overall, they persuade through emotion and shared experience, making debates lively, enjoyable, and easy for others to participate in.
ENFP	In debates, ENFPs are enthusiastic and highly creative communicators. They are skilled at injecting fresh ideas into discussions and often introduce connections and perspectives that others would not have considered. ENFPs enjoy presenting their arguments through compelling stories, analogies, and emotional resonance, making their points both logical and heartfelt. Their style is upbeat and slightly playful, bringing energy to the debate. ENFPs are typically open-minded, eager to hear different opinions, and often adjust their positions flexibly as they absorb new ideas. However, they may sometimes drift off topic due to excitement or become distracted by intriguing thoughts; and because they invest deeply in the discussion emotionally, sharp opposition can momentarily trigger emotional reactions. Even so, their optimism and people-centered approach often help diffuse tension and encourage everyone to seek common ground.
ENTP	In debates, ENTPs can be described as natural-born debaters, treating debates as intellectual games through which they challenge both others and themselves. Quick-thinking and outspoken, ENTPs are hardly ever afraid to disagree with anyone or any viewpoint; in fact, they become energized when presenting alternative perspectives. They enjoy playing the "devil's advocate," sometimes even defending positions they do not personally believe in, simply to uncover their flaws and savor the thrill of debate. Their communication style is witty and sharp, often laced with humor and sarcasm-disarming opponents while simultaneously exposing weaknesses in their arguments. ENTPs thrive in fast-paced, spontaneous exchanges, frequently delivering clever rebuttals that catch others off guard. However, this improvisational agility also means they may jump between multiple points or get sidetracked by tangents.

nouns, verbs, adjectives, and adverbs) to the total number of tokens in a text, serving as a proxy for informational richness and conceptual compactness of the prompt.

For measuring personality-question similarity, we employ the *all-mpnet-base-v2*[4] sentence embedding model. Specifically, both the personality-pair descriptions and the full question texts (including answer options) are encoded into dense vector representations, and cosine similarity between these embeddings is used as the similarity score.

To automatically select effective personality pairs and perform domain classification, we fine-tune a *RoBERTa-base* model [9]. The model is trained with a maximum sequence length of 512, a batch size of 8, and a learning rate

[4] https://huggingface.co/sentence-transformers/all-mpnet-base-v2.

Table B.5
Examples of debate-behavioral personality descriptions adapted from the MBTI framework (Part 4 of 4).

MBTI Personality	MBTI-Debate
ESTJ	In debates, ESTJs display the demeanor of leaders and a strong respect for order. They often take the initiative to structure the discussion, ensuring that the debate proceeds in an organized manner and stays focused on the topic. When presenting their viewpoints, ESTJs are direct and fact-based, supporting their positions with evidence and established rules, and they are adept at citing past experience or authoritative sources as reinforcement. They do not hesitate to point out errors or contradictions in an opponent's argument, expecting the debate to operate within a clear and logical framework. If the discussion strays off topic or becomes inefficient, ESTJs will decisively steer it back on track. Because they are assertive and firm in their convictions, ESTJs may sometimes unintentionally dominate the conversation, making others feel that they are somewhat overbearing.
ESFJ	In debates, ESFJs are friendly and focused on creating a harmonious atmosphere. They tend to adopt a cooperative stance, often acknowledging the reasonable parts of the other side's arguments and trying to build common ground. When presenting their viewpoints, ESFJs draw heavily on everyday examples and emphasize the impact on others or the community to gain emotional resonance from the audience. They are highly sensitive to tension in discussions; whenever they sense rising conflict, they often step in as mediators to ease the situation. When refuting others, ESFJs use gentle language and try to avoid direct offense-even when they disagree, they express their objections tactfully. They care deeply about others' reactions and feelings, and may feel hurt if their opinions are harshly criticized. They also tend to avoid conflict topics that could cause discomfort.
ENFJ	In debates, ENFJs display a charismatic and inspiring presence. They often step into the role of a motivator, uniting the audience with their enthusiasm and sincerity. ENFJs have a keen understanding of human nature and use compelling stories or shared values to make their arguments resonate deeply with listeners. Their communication style is warm, passionate, and inclusive, frequently encouraging and rallying others to stand with them. ENFJs are also attentive listeners: if someone feels overlooked or emotionally affected during the discussion, they will take the initiative to acknowledge and comfort that person, ensuring everyone remains engaged. On the other hand, ENFJs can be somewhat idealistic, and if their efforts to build consensus fail, they may feel internally discouraged. Additionally, because they value others' approval and feedback, sharp opposition can affect them emotionally even if they appear composed on the surface.
ENTJ	In debates, ENTJs exhibit a forceful and decisive style. They often treat debates as battles to be won, coming in well-prepared, armed with clear arguments and a confident offensive-and-defensive strategy. When presenting their viewpoints, ENTJs speak with confidence and conviction, naturally taking control of the discussion and steering it toward conclusions that favor their position. They place great emphasis on logic and efficiency, quickly and directly dismantling any weak points in their opponent's arguments. Their rebuttals are sharp and to the point; they do not hesitate to use cutting language, as winning the debate is their top priority. At times, ENTJs can dominate the conversation so strongly that their opponents have little chance to speak. If they become aware of this tendency, they may restrain themselves enough to allow others to express their views-yet by instinct, they still gravitate toward leading the entire debate.

of 2×10^{-5}. Training follows a dynamic early-stopping strategy in which optimization terminates when the validation loss reaches its minimum, preventing overfitting and promoting stable generalization.

All benchmark datasets used in our experiments (e.g., MMLU-Pro) are publicly available and are used in accordance with their original licenses and terms of use. Moreover, all experiments are conducted on a server equipped with eight NVIDIA A100 GPUs, running Ubuntu 22.04.4 LTS.

Table B.6
Examples of rewritten direct personality formulations for single-turn question answering (Part 1 of 2).

MBTI Personality	MBTI-Direct
ISTJ	ISTJs are detail-oriented, logical, and methodical problem solvers. In standardized tests, they rely on factual recall and step-by-step elimination of wrong answers, preferring accuracy over speed and sticking to proven methods. In decision-making, they focus on concrete data and past experiences, making pragmatic choices with little sway from emotions. In logic puzzles, they apply structured analysis and patience, systematically working through the problem rather than seeking creative leaps.
ISFJ	ISFJs are meticulous and supportive problem solvers who balance practical detail-focus with concern for others. In exams, they double-check answers and value thorough accuracy. In decisions, they rely on proven experience and seek harmonious outcomes. In puzzles, they follow step-by-step methods and ensure no detail is overlooked, though they may avoid overly abstract approaches.
INFJ	INFJs are insightful, future-oriented thinkers who search for hidden patterns and deeper meaning. In tests, they often read between the lines and seek conceptual answers, sometimes overthinking simple questions. In decisions, they aim for solutions aligned with values and long-term welfare. In puzzles, they rely on big-picture insight and creative reasoning, though they may consider too many possibilities.
INTJ	INTJs are strategic and independent problem solvers who identify underlying principles and long-term implications. In tests, they apply efficient strategies to maximize accuracy and speed. In decisions, they use objective pros-and-cons analysis and future-oriented reasoning. In puzzles, they combine systematic planning with innovative thinking to reach optimal solutions.
ISTP	ISTPs are analytical and adaptable problem solvers who focus on concrete facts and logical principles. In exams, they break questions down logically and perform well under time pressure. In decisions, they remain pragmatic and objective. In puzzles, they use trial-and-error combined with sharp reasoning to rapidly adjust toward effective solutions.
ISFP	ISFPs are gentle and observant problem solvers guided by personal values. In exams, they rely on memory and intuitive resonance, preferring accuracy over speed. In decisions, they choose options aligned with values and present needs. In puzzles, they work step by step using concrete clues, often avoiding overly abstract logic.
INFP	INFPs are idealistic and imaginative solvers who prioritize values and empathy. In tests, they may follow intuition and reflect deeply on meaning, sometimes sacrificing speed. In decisions, they seek outcomes aligned with moral vision. In puzzles, they explore patterns creatively but may overthink or miss small details.
INTP	INTPs are logical and inventive thinkers who enjoy analyzing complex ideas. In tests, they examine all angles thoroughly, sometimes slowing down due to overanalysis. In decisions, they delay choices until the most rational option is clear. In puzzles, they discover creative solutions but risk overcomplicating tasks.

E. Effects of Personality-Pair Conditioning on MAD Performance

We fit a mixed-effects regression model that uses the added personality pairs to predict inference accuracy, while controlling for a random effect for each model. The control setting, in which no personality is added to the system prompt, is used as the reference category. Figure E.1 presents the aggregated results across all evaluated models, ranking all MBTI personality pairs by their effect sizes relative to the no-personality control setting. Figures E.2-E.11 further show the corresponding model-specific results for each of the ten individual models.

From these results, we observe substantial variation in model sensitivity to personality-pair conditioning. For example, Qwen3-8B exhibits a much larger number of personality pairs with positive effects, whereas smaller models such as Qwen3-0.6B show substantially fewer beneficial personality pairs.

Table B.7
Examples of rewritten direct personality formulations for single-turn question answering (Part 2 of 2).

MBTI Personality	MBTI-Direct
ESTP	ESTPs are pragmatic and action-oriented problem solvers. In exams, they prioritize speed and concrete elimination of wrong answers. In decisions, they address immediate facts efficiently and adapt quickly if conditions change. In puzzles, they rely on trial-and-error and thrive under pressure, though they may overlook deeper patterns.
ESFP	ESFPs are outgoing, hands-on problem solvers who favor practical approaches and teamwork. In exams, they use common sense and experience, making educated guesses if needed. In decisions, they focus on immediate happiness and adaptability. In puzzles, they combine tangible clues with enthusiasm and creativity to find effective solutions.
ENFP	ENFPs are imaginative problem solvers who see many possibilities. In tests, they rely on flashes of insight but must guard against overlooking details. In decisions, they generate creative options guided by values. In puzzles, they excel at novel solutions but may need to rein in ideas to focus on accuracy.
ENTP	ENTPs are inventive and quick-witted problem solvers who enjoy mental challenges. In exams, they spot patterns and use creative shortcuts, though they may skim instructions. In decisions, they brainstorm and pivot flexibly. In puzzles, they combine strategic logic with originality, sometimes skipping simpler methods.
ESTJ	ESTJs are efficient, results-driven problem solvers who rely on proven methods. In exams, they work systematically with strong time management. In decisions, they choose practical options quickly and confidently. In puzzles, they apply step-by-step strategies and rarely deviate from established rules.
ESFJ	ESFJs are conscientious and cooperative problem solvers focused on accuracy and harmony. In exams, they follow structured strategies and double-check answers. In decisions, they seek consensus and precedent. In puzzles, their patience and attention to detail help them reach correct solutions, especially in practical contexts.
ENFJ	ENFJs are empathetic and visionary problem solvers who consider human impact and group welfare. In exams, they combine preparation with intuition. In decisions, they act decisively once values are aligned. In puzzles, they blend big-picture insight with organized effort, often seeking purposeful outcomes.
ENTJ	ENTJs are decisive and strategic problem solvers who pursue efficiency and results. In exams, they quickly identify frameworks and eliminate weak options. In decisions, they apply cost-benefit logic and long-term strategy. In puzzles, they break problems into parts and solve them systematically, often competing against time.

F. Domain-Level Effects of Personality-Pair Conditioning on MAD Performance

We fit a mixed-effects regression model that uses the added personality pairs to predict inference accuracy within each domain, while controlling for a random effect for each model. The control setting, in which no personality is added to the system prompt, is used as the reference category. Figures F.1-F.10 present the resulting domain-level effects for each of the ten evaluated models.

Consistent with the overall performance trends, we observe substantial variability across models. For example, Qwen3-8B exhibits significant performance improvements in seven domains, including Business, Math, and Physics, whereas Qwen2.5-7B shows a significant improvement only in the Math domain. Moreover, across models, personality-pair conditioning tends to yield positive effects primarily in the Math and Business domains, while consistently leading to performance degradation in domains such as Biology, History, and Psychology.

G. Mixed-Effects Analysis of Factors Influencing Personality-Driven Debate

To conduct a comprehensive analysis of all previously discussed personality-related attributes, we fit a mixed-effects regression model with three fixed effects: the lexical density of personality descriptions, the similarity between personality descriptions and questions, and prompt perplexity. Random intercepts are included for each model. Table G.1 reports the regression results. We find that higher lexical density, lower personality-question similarity,

Table C.1
Selected MBTI personalities for each domain using the GLM-4-9B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	INFJ	ISTJ	ESFP	INFP	ESTJ	ENTJ
Business	INTP	ENFJ	ENTP	ISTP	ENTJ	ESFJ
Chemistry	ENTP	ESTJ	ENFJ	ISTJ	ISTP	INFP
Computer Science	INTJ	ISFJ	INFJ	INTP	ESTJ	ENTJ
Economics	ISFJ	ISFP	ENFJ	INFJ	ENFP	ENTP
Engineering	ENTJ	ENFJ	ISFJ	ISFP	ESTJ	ENTP
Health	INTP	ENTP	INTJ	ESTP	ISTP	ESFP
History	INTJ	ISFJ	INTP	ESFP	INFP	ESTP
Law	INFJ	INTJ	ESTP	ENFJ	ENTP	INFP
Math	INTP	ISFJ	ISTP	ISFP	ENTJ	ESTJ
Other	INTJ	ISFJ	INTP	ESFP	ENTP	ESTP
Philosophy	ISFP	INFP	ISTP	ENFP	ESFJ	ENFJ
Physics	ISTJ	ISTP	ESFP	INTP	ESFJ	ESTJ
Psychology	ENFP	ESFP	ISTP	INTP	ESTJ	ENTJ

Table C.2
Selected MBTI personalities for each domain using the LLaMA-3.1-8B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ISTP	ENTP	INTJ	INFP	ESFP	INFJ
Business	ENTJ	ISFP	ESTJ	ESFJ	ENFP	ESFP
Chemistry	ISTP	INFP	ENTP	ESFJ	ISFJ	ESTJ
Computer Science	ISTP	INTP	ESTJ	ENTJ	ENTP	ENFP
Economics	ESFP	ESFJ	ESTJ	ISTJ	INTJ	INTP
Engineering	INTP	ENTJ	ESFJ	ISFP	ESTJ	ISFJ
Health	ISTP	ISFP	ISFJ	ENFP	ENFJ	ISTJ
History	ISTP	ISTJ	INTJ	ISFP	ENFP	INFP
Law	ENFJ	INFJ	ISTJ	INFP	ESTJ	INTP
Math	ENFP	INFJ	ESTJ	INTJ	ISTJ	ISTP
Other	ISFJ	ESTJ	INTJ	INFP	ESFP	ENTP
Philosophy	ESTJ	ISFJ	INFJ	INTJ	ENFP	ENFJ
Physics	ISFJ	ISTJ	ESTP	INFJ	ENFJ	INFP
Psychology	ISTP	ISFP	ENFJ	ESFP	ESTP	ENTJ

and lower prompt perplexity are generally associated with higher prediction accuracy ($p < 0.05$), consistent with the trends observed in the correlation analysis.

H. Model Consistency of MBTI Personality-Pair Effects on MAD Performance

Figure H.1 illustrates the correlations between the mean accuracy of personality pairs across 700 questions and their performance across four representative models. Among all model pairs, LLaMA-3.1-70B and Qwen2.5-32B exhibit the highest correlation (0.51), whereas GLM-4-9B and Qwen2.5-32B show the lowest correlation (0.29).

I. Human Evaluation of Personality Adherence

To assess whether models adhere to the intended personality descriptions during debate, we conduct a human evaluation using a *Personality Adherence Score (PAS)*. We recruit three professional annotators to evaluate the extent to which each model's debate responses conform to the predefined MBTI personality traits. For each of the ten LLMs, we randomly sample 30 debate instances, which are independently rated by all annotators on a five-point Likert scale (1-5), where higher scores indicate stronger adherence to the target personality. The raw scores are then linearly normalized

Table C.3
Selected MBTI personalities for each domain using the LLaMA-3.1-70B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	INTP	ISTJ	INFP	ESFP	ENTP	ENFP
Business	ENFP	ISFJ	ESFJ	ISTJ	ISTP	ESTJ
Chemistry	ESFP	ESTP	ENFJ	INTJ	ISTJ	ESTJ
Computer Science	INFP	INTP	ENTP	INFJ	ISTJ	ESTJ
Economics	ISTP	ESFP	INTP	ISFP	INTJ	ENTJ
Engineering	ISFJ	ESFJ	ENFJ	ESTP	ESTJ	ISTP
Health	ISTJ	ISFJ	ENFP	ESTJ	ISTP	ESTP
History	ISTP	ISTJ	INFP	ESFP	ENFJ	ENTJ
Law	ISFP	ISFJ	ENFJ	ISTJ	ENFP	ESTJ
Math	ENTP	ISFP	ENTJ	ISTJ	ESTP	ENFJ
Other	ESTP	ISFP	ENTP	ISFJ	ESFJ	ESTJ
Philosophy	ISTJ	ESTJ	ESFP	ENFP	ESTP	ESFJ
Physics	ISTP	ISTJ	ENTP	ESFJ	ESTP	INFP
Psychology	ISTJ	ESTJ	INFJ	ENFJ	ENTP	ENFP

Table C.4
Selected MBTI personalities for each domain using the Qwen2.5-7B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ENFJ	INTP	ISFP	ENTJ	INFJ	ISFJ
Business	INTJ	INTP	ENFJ	ISFP	ENTJ	INFP
Chemistry	ISTJ	ISFP	ENFJ	ESFJ	ENTP	ENTJ
Computer Science	ENTJ	INTJ	ENTP	INFJ	ISFP	ISFJ
Economics	INTP	ISTP	ENFP	ESTJ	ESTP	ESFP
Engineering	INTP	ESTJ	ESFP	ESFJ	INFP	INFJ
Health	ISTP	ISTJ	ENFJ	ESFP	INFJ	INFP
History	ISFP	ISFJ	ISTP	ESTJ	INTJ	ENTJ
Law	ENFP	ENFJ	ESFJ	ISTP	ISFP	ISFJ
Math	INFP	ENFP	INFJ	ENTP	ESTJ	ISTJ
Other	ISTJ	ENFJ	INFP	ESFP	ENFP	ENTP
Philosophy	ENTP	INTP	INFJ	ESFJ	ESTJ	ISFP
Physics	ISFP	ENTJ	INTP	ESFP	ESTJ	ENFJ
Psychology	ISFJ	ESTJ	INFJ	ESFP	ISFP	INFP

to the range [0, 1], and the resulting personality adherence scores are summarized in Table I.1. Annotating each sample requires between 10 and 20 minutes, with an average annotation time of 13 minutes per sample. In accordance with local labor regulations, annotators are compensated at a rate of $3.15 per hour. We assess inter-annotator agreement using Fleiss' κ, obtaining a value of 0.809, which indicates a high level of consistency among annotators. The full annotation instructions are publicly available in our anonymous repository.

From Table I.1, we observe substantial variation across models in their adherence to the intended personality-based debate behaviors. Larger models generally achieve higher PAS scores, indicating more consistent expression of the prescribed personality traits. For example, LLaMA-3.1-70B and Qwen3-8B attain PAS scores of 0.658 and 0.801, respectively, whereas smaller models exhibit noticeably weaker adherence, with Qwen3-0.6B achieving a PAS score of only 0.181.

Inspired by classical reliability theory and established interpretations of agreement coefficients [7, 11, 4], we assess whether models reliably instantiate the intended personalities using a fixed PAS threshold of 0.40. Prior work commonly interprets agreement values below 0.40 as indicating at most fair reliability, which is generally considered insufficient for stable and interpretable behavioral assessment, whereas higher values correspond to more reliable and consistent patterns. Following this guidance, we exclude models whose PAS falls below this threshold from our main

Table C.5
Selected MBTI personalities for each domain using the Qwen2.5-14B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ENTP	INTP	ISFP	ENFP	ESFJ	ESFP
Business	ISFP	ESTP	INTJ	ISTP	ISTJ	INFJ
Chemistry	ESFP	ENFP	INFP	ISTJ	ENTP	ISTP
Computer Science	INTJ	INTP	INFP	ESTP	ISFJ	ESFJ
Economics	ESTP	ENTP	INFP	INFJ	ISTJ	ISFJ
Engineering	INTP	INFJ	ISTJ	ISFP	ESFP	ESTP
Health	ENTP	ISFP	ESTJ	INFP	ENFP	ESFJ
History	INTJ	ESTJ	ENTP	ENFJ	ESFP	ISTJ
Law	INTP	ENTJ	ISTJ	INFJ	ISFP	ESFJ
Math	ENTP	ENTJ	INTJ	INFJ	ESFJ	ISFJ
Other	ENFP	INFP	ISTP	ESFP	ESTJ	ESFJ
Philosophy	ESTJ	INTJ	ISFP	INTP	INFP	ENFP
Physics	INFP	ENTP	ENTJ	ISFJ	ESFP	ISTJ
Psychology	ISTP	ESFP	INTJ	ENTJ	ESFJ	INFJ

Table C.6
Selected MBTI personalities for each domain using the Qwen2.5-32B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ENTJ	INTP	INFP	ESFP	ISTP	ESFJ
Business	ENTP	ESFJ	INFP	ENFP	ESTJ	ISFP
Chemistry	INFJ	ENFP	INTJ	ISTP	ESTJ	ENTJ
Computer Science	INFP	ESTJ	ESTP	ENTP	ISTP	ISFP
Economics	ESFJ	ISFJ	ENFP	ISTP	ENTJ	INTJ
Engineering	INTJ	INTP	ISTP	INFP	ISFJ	ISFP
Health	ESTJ	ISFJ	INFP	ENFJ	ENFP	ENTP
History	ENTP	INTP	INFJ	ENTJ	ISTJ	ESTJ
Law	ENTJ	INTJ	ESTP	ISTP	INFP	ISFP
Math	INTP	INFJ	INTJ	ESFP	ENTJ	ESTJ
Other	ESTP	ESFP	ESTJ	INFJ	ESFJ	ENFJ
Philosophy	ISTJ	ESTJ	ENTJ	INTJ	ESFJ	ENFJ
Physics	INTJ	INTP	ISTP	INFP	ISFP	ESFP
Psychology	ISFJ	ISTJ	INTP	ESFJ	INFJ	INFP

experiments (Figures E.1), ensuring that personality-driven effects are analyzed only for models with minimally reliable personality adherence.

J. More Detailed Single-Agent Evaluation

Table J.1 summarizes the performance of the three MBTI personality prompts across different backbone models, with accuracy averaged over three prompting strategies (MBTI-Orig, MBTI-Debate, and MBTI-Direct). The *Avg.* row reports the mean accuracy across all three models. Notably, certain MBTI personality types, such as ISTJ and INTJ, show consistent performance improvements across all evaluated models.

K. Illustrative Debate Examples

Tables K.1 and K.2 present two representative debate cases with full interaction transcripts. In Case 1, introducing personality-conditioned agents leads to a change in the debate outcome: the ENTP agent is convinced by the ENTJ agent, and the discussion converges on the correct answer. In contrast, in the personality-neutral setting, the agent advocating the correct option is persuaded by the opponent, resulting in an incorrect final decision. A different pattern

Table C.7
Selected MBTI personalities for each domain using the Qwen3-0.6B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Business	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Chemistry	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Computer Science	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Economics	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Engineering	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Health	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
History	ISTJ	ISFJ	ESFP	ENFP	INTP	ESTP
Law	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Math	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Other	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Philosophy	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Physics	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ
Psychology	ISTJ	ISFJ	INTP	ESTP	ENFJ	ENTJ

Table C.8
Selected MBTI personalities for each domain using the Qwen3-1.7B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ENFJ	INFJ	ENFP	ESTP	ISTJ	ESTJ
Business	INFJ	INFP	ENTP	ESTJ	ISTP	ISTJ
Chemistry	INFJ	INTP	INFP	ESTP	ESFP	ESTJ
Computer Science	ESFJ	ENFP	ISFP	INFJ	ISTJ	ISTP
Economics	ENTP	ISFP	INTJ	ESTJ	ESFP	ISTJ
Engineering	ESTP	ENFJ	INTJ	ISTP	ISFJ	ISTJ
Health	ESTJ	ENFJ	INTJ	ENTP	ESFP	ISFJ
History	ENFJ	ISFP	ENTJ	INFJ	INFP	ENTP
Law	ESTP	ISTJ	ENTP	ISTP	ENTJ	ISFP
Math	INFJ	ENFJ	ISTJ	INTP	ESTP	ESFP
Other	ESTJ	INTJ	ENFJ	ISFP	INFP	ENFP
Philosophy	ESFJ	ISFJ	ESTJ	ISTP	ENTP	INTP
Physics	ESFJ	ESTP	ESTJ	ISTP	ENTJ	ENTP
Psychology	ISTJ	ISTP	ESFJ	ENTJ	INFP	ENFP

emerges in Case 2. In the personality-conditioned debate, the INTP agent is convinced by the ENTP agent, causing the debate to converge on an incorrect answer. By contrast, in the personality-neutral debate, the agent defending the correct option successfully persuades the opponent, leading to the correct outcome. Together, these two cases illustrate that incorporating personality into debate agents can substantially reshape persuasion dynamics and final conclusions, sometimes positively and sometimes negatively, depending on how different reasoning styles interact.

L. Effects of Pro&Con Role Swapping in Personality-Pair Debates

To examine how swapping proponent and opponent roles affects MAD performance for MBTI personality pairs, we compute, for each personality pair, the accuracy difference between a given proponent-opponent assignment and its reversed role assignment, and then average these differences across models. Figure L.1 shows the results aggregated across all evaluated models, highlighting how role swapping influences debate performance for different personality pairs. Figures L.2-L.11 present the corresponding heatmaps for each of the ten individual models.

In each heatmap, the lower-left and upper-right triangles exhibit symmetric values with opposite signs, reflecting the fact that swapping the proponent and opponent roles reverses the direction of the measured performance difference. From Figure L.1, we observe a clear systematic directional bias in personality-pair effects across debate roles: certain

Table C.9
Selected MBTI personalities for each domain using the Qwen3-4B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	ISTJ	ISFJ	INTP	ISFP	ENTP	ESTP
Business	ESFP	ESFJ	ENFP	ENFJ	ENTP	ISTJ
Chemistry	ESFP	ENFP	INFP	ESTP	ESTJ	ESFJ
Computer Science	ENFP	ISFP	INFJ	ENTP	ISFJ	ISTJ
Economics	ISFP	ENFJ	INFJ	ENTJ	ESTJ	ISTJ
Engineering	ENFP	ENTP	INTP	ESFP	ISTJ	ESTJ
Health	ENFP	ENTJ	INFJ	INTP	ESFJ	ESTP
History	ESTJ	ISTJ	ESFP	ENTP	ESTP	ISTP
Law	ISFJ	ESTP	ESTJ	INTP	INFJ	ENFJ
Math	INTP	INFJ	ISTP	ISFP	ISFJ	INTJ
Other	ENFJ	ESTJ	INFP	ESFJ	ESFP	ENFP
Philosophy	INTP	ESTJ	ESTP	ISTP	INFP	ENFP
Physics	ENFP	INTJ	ISFP	ESTP	ENFJ	ESTJ
Psychology	ESTJ	ISTJ	ENFJ	ISFJ	ISTP	ENFP

Table C.10
Selected MBTI personalities for each domain using the Qwen3-8B model.

Domain	MBTI-1	MBTI-2	MBTI-3	MBTI-4	MBTI-5	MBTI-6
Biology	INTP	ESTJ	ENTJ	ISFJ	ENFJ	ENTP
Business	ISTJ	INTJ	ENTJ	ISFJ	ISFP	ENFJ
Chemistry	ISTJ	ISTP	ISFP	ENFP	ESTP	ENTP
Computer Science	INTJ	ISTJ	INFP	INFJ	ENFP	ENFJ
Economics	ESTJ	ENTJ	ISFP	ESTP	ENTP	ENFJ
Engineering	ISTJ	INTJ	ENFJ	INFP	ESFP	ESTP
Health	INTP	ESTJ	ESFJ	ISFJ	ESFP	ENFP
History	INTJ	ISTP	INTP	ENFP	ESFP	ENFJ
Law	ESTJ	ESTP	ISFP	ISTJ	ESFJ	ENFJ
Math	ISTJ	INFJ	ISFP	INTP	ESFP	ESFJ
Other	ENTJ	INTP	ISFJ	ISFP	ESTJ	ESFJ
Philosophy	ESTJ	ISTJ	ENTP	ENFJ	ENFP	ESFP
Physics	ESTJ	INTP	ISFP	INFP	ESFP	ESTP
Psychology	ISTJ	INTJ	ESTP	ISFJ	ENTP	ENFJ

Table G.1
Coefficients of the mixed-effects model capturing the relationship between inference accuracy and personality attributes.

Term	**Coefficient**	**p-value**
Lexical_density	0.072514375	4.634e-29
Similarity	-0.094005563	1.232e-42
Perplexity	-0.273095766	<1e-320

personalities consistently yield better performance when assigned to the proponent role, while others are more effective as opponents. For example, swapping the roles of the INTJ-ESFP personality pair results in a performance difference of 0.235, whereas the INFP-ENTJ pair exhibits a performance difference of -0.218. Moreover, the magnitude and direction of these role-swapping effects vary substantially across models. For instance, in GLM-4-9B (Figure L.2), swapping the ISTP-ENTJ personality pair leads to a performance decrease of -0.223, while in LLaMA-3.1-8B (Figure L.3), the same role swap yields a positive performance difference of 0.105.

Model	PAS ↑
GLM-4-9B	0.372
LLaMA-3.1-8B	0.461
LLaMA-3.1-70B	0.658
Qwen2.5-7B	0.458
Qwen2.5-14B	0.508
Qwen2.5-32B	0.703
Qwen3-0.6B	0.181
Qwen3-1.7B	0.367
Qwen3-4B	0.581
Qwen3-8B	**0.801**

Table I.1
Human-evaluated Personality Adherence Scores (PAS) for each model, where higher scores indicate stronger consistency between model debate behaviors and the intended personality descriptions.

Table J.1
Performance comparison of the top three MBTI personality prompts (ISTJ, INTJ, and ISTP) across three models. Results are averaged over three MBTI prompts. *Avg.* denotes the average accuracy across the three models, with higher values indicating better performance.

Model	ISTJ↑	INTJ↑	ISTP↑	Single_Agent
Qwen3-8B	**0.4424**	0.4419	0.4371	0.3629
GLM-4-9B	0.4305	**0.4348**	0.4181	0.3443
LLaMA-3.1-70B	**0.4871**	0.4833	**0.4871**	0.4271
Avg.	**0.4533**	**0.4533**	0.4474	0.3781

M. Prompt Templates for MBTI-Driven MAD

Table M.1 details the system, initial, iterative, and judgment prompts used for the proponent, opponent, and moderator roles, illustrating the complete debate workflow and interactions among roles. Notably, the moderator operates independently from the debating agents throughout the debate process. It is designed to be strictly neutral: the moderator does not exhibit any personality traits and does not introduce new evidence. Instead, it aggregates and evaluates the arguments presented by both sides and selects the most plausible option.

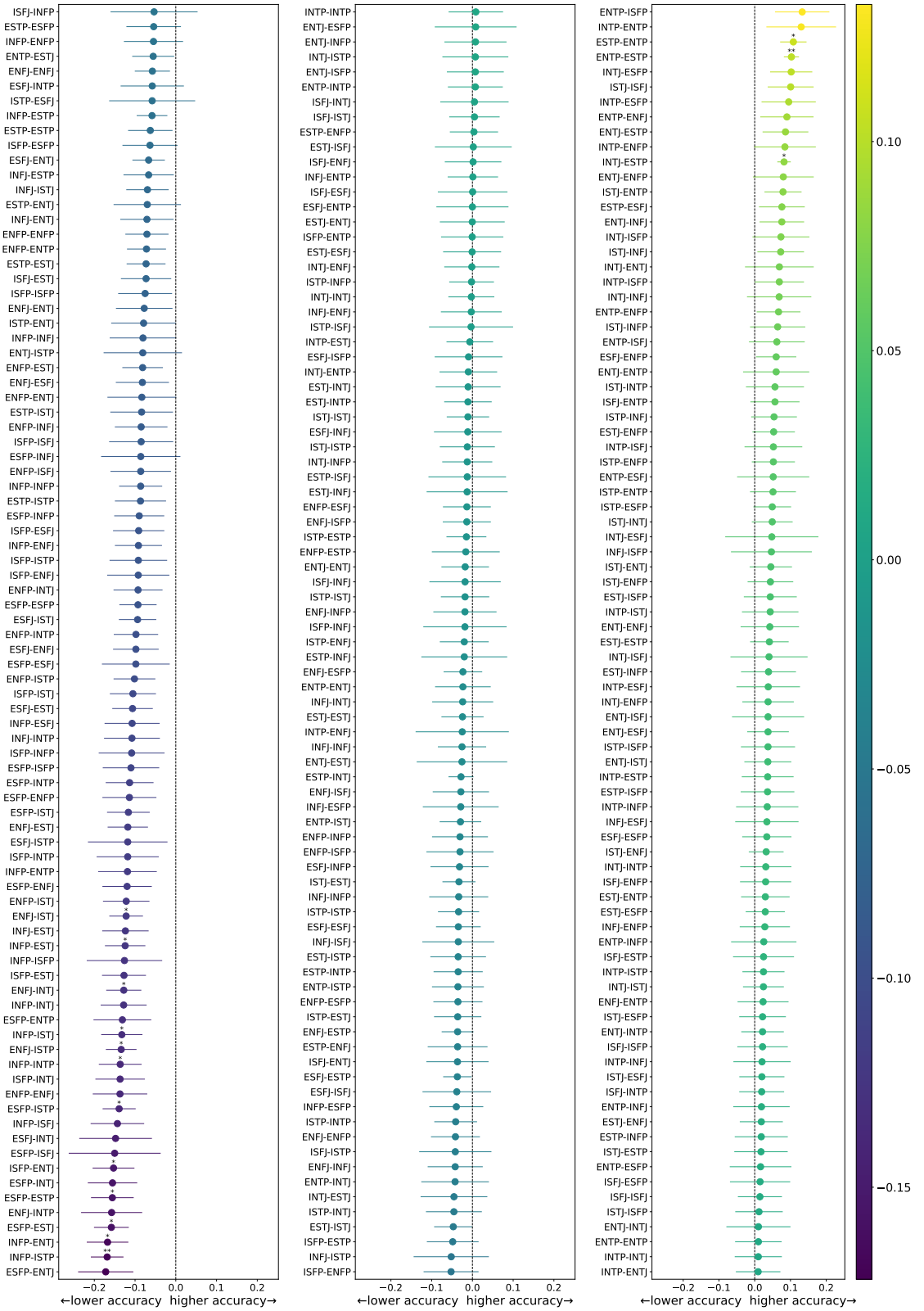

Figure E.1: All MBTI personality pairs ranked by their effect sizes on performance change relative to the control setting, with models exhibiting low PAS excluded from the analysis.

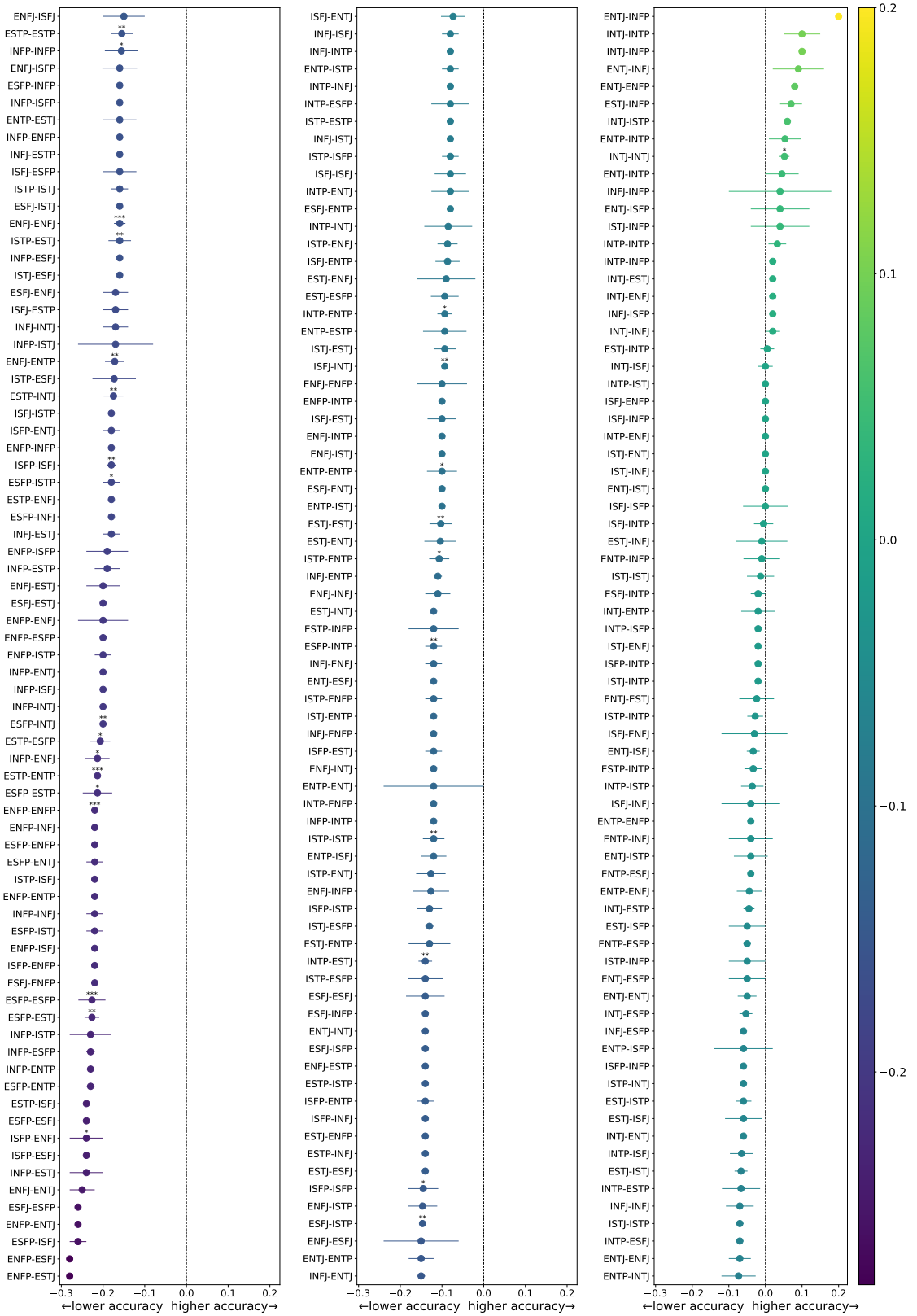

Figure E.2: All MBTI personality pairs ranked by their effect sizes on performance change for GLM-4-9B relative to the control setting.

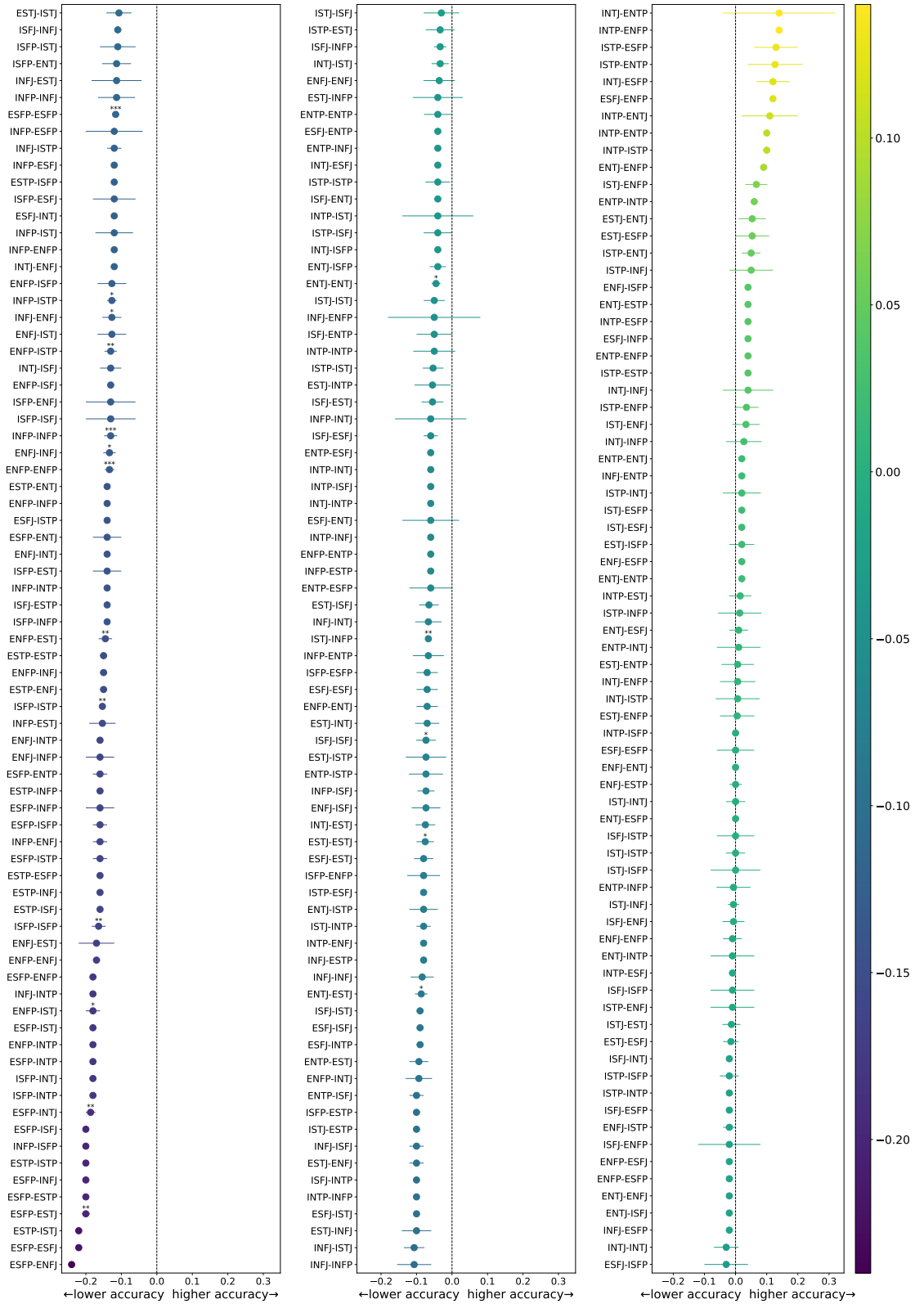

Figure E.3: All MBTI personality pairs ranked by their effect sizes on performance change for LLaMA-3.1-8B relative to the control setting.

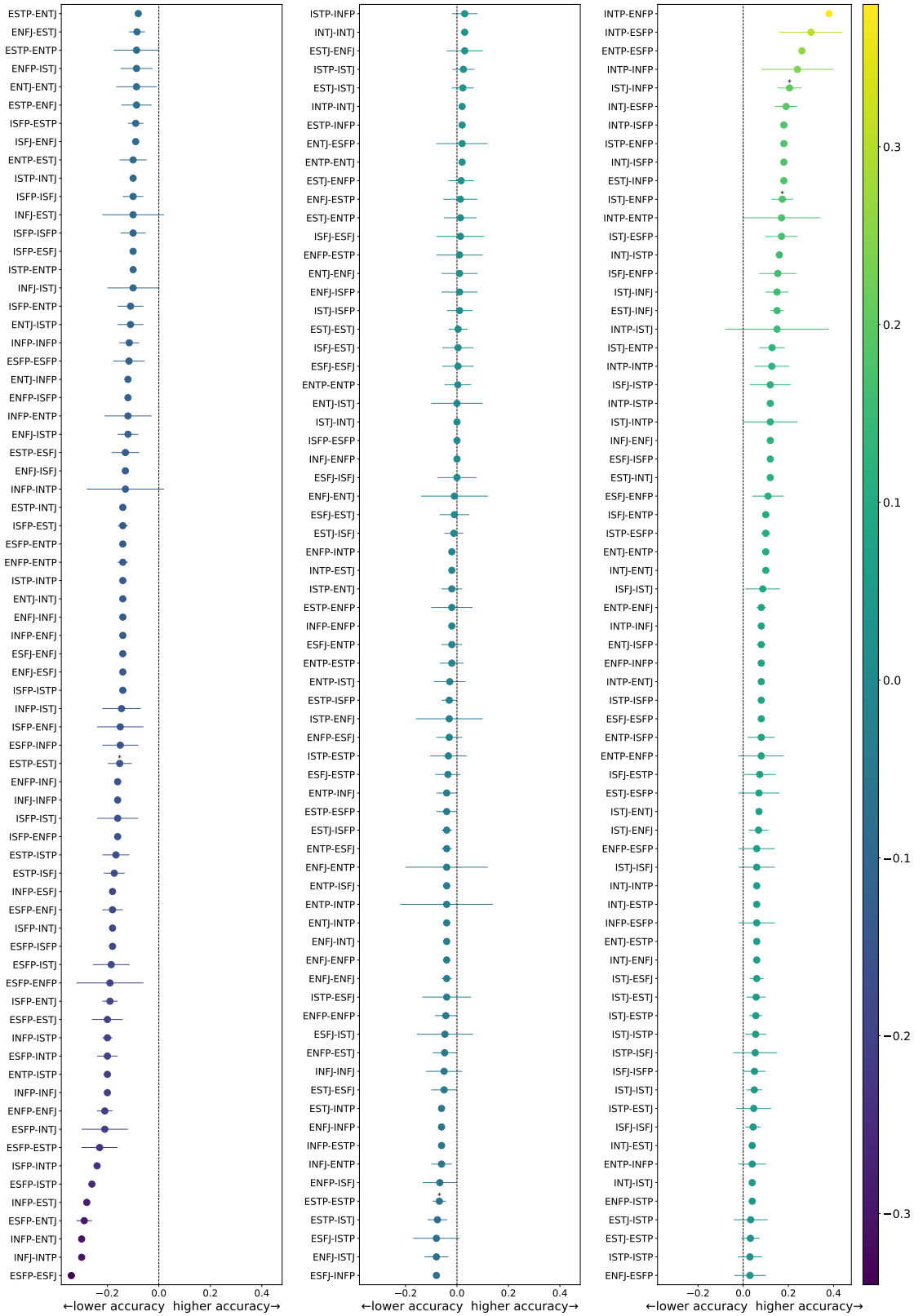

Figure E.4: All MBTI personality pairs ranked by their effect sizes on performance change for LLaMA-3.1-70B relative to the control setting.

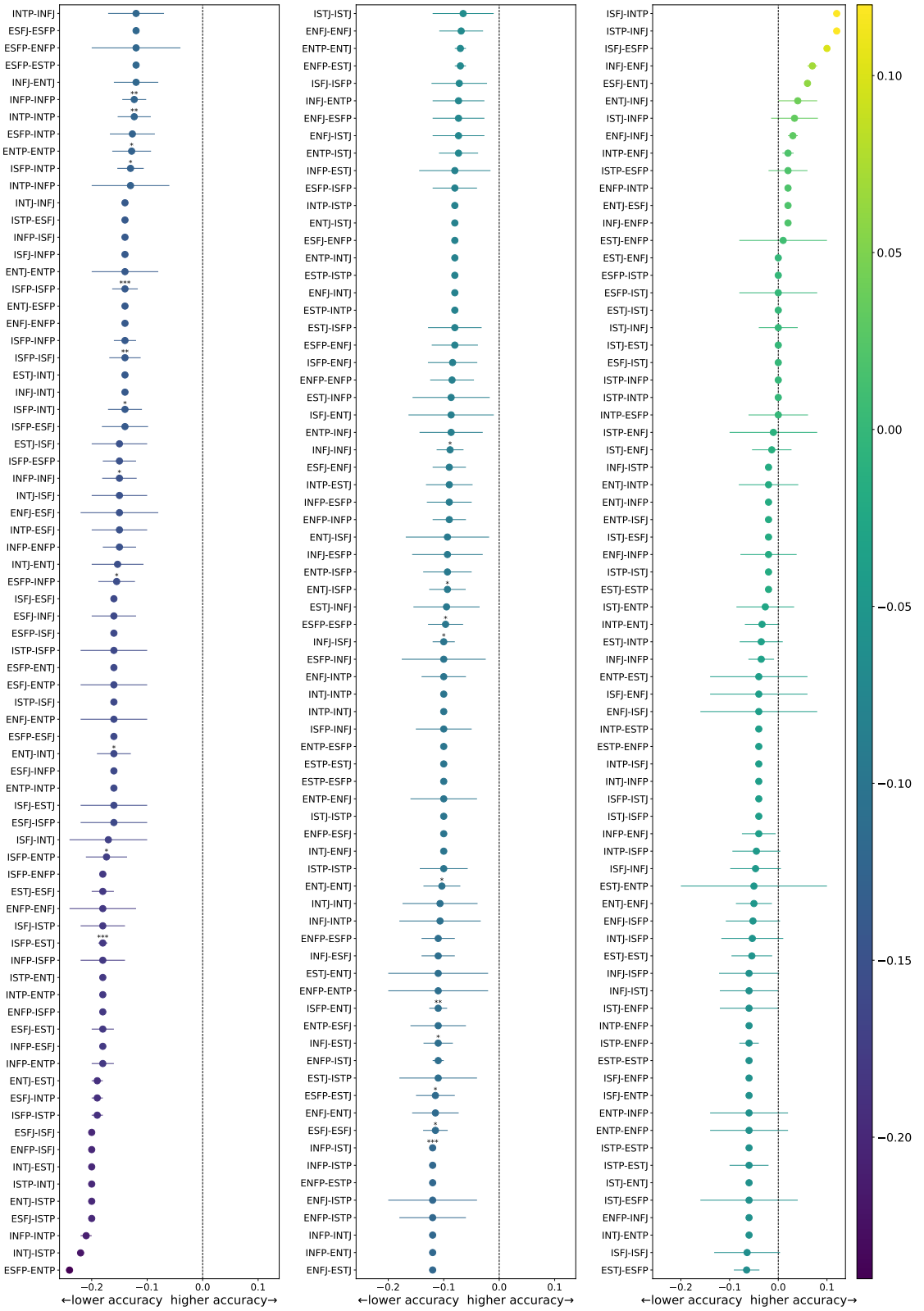

Figure E.5: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen2.5-7B relative to the control setting.

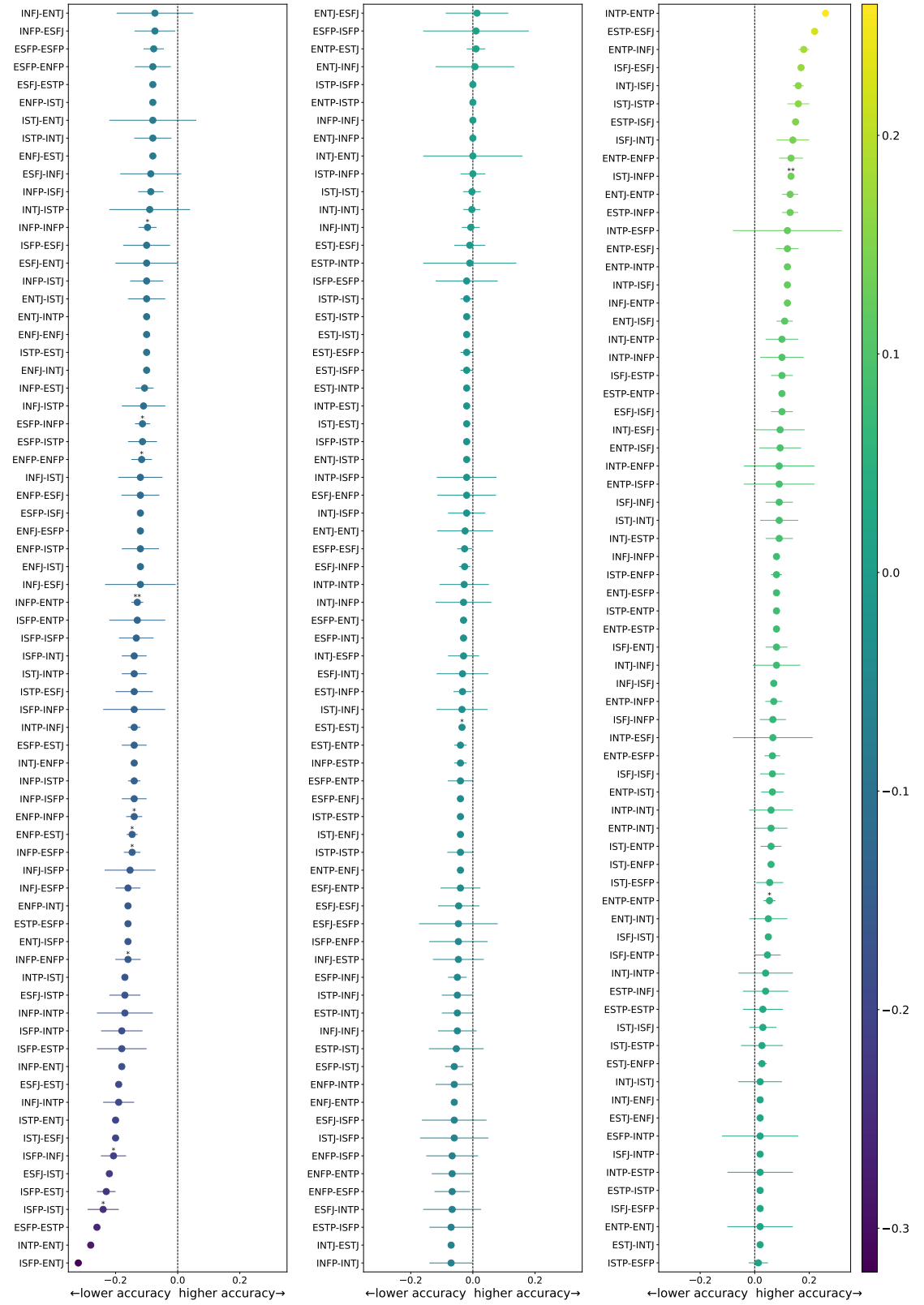

Figure E.6: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen2.5-14B relative to the control setting.

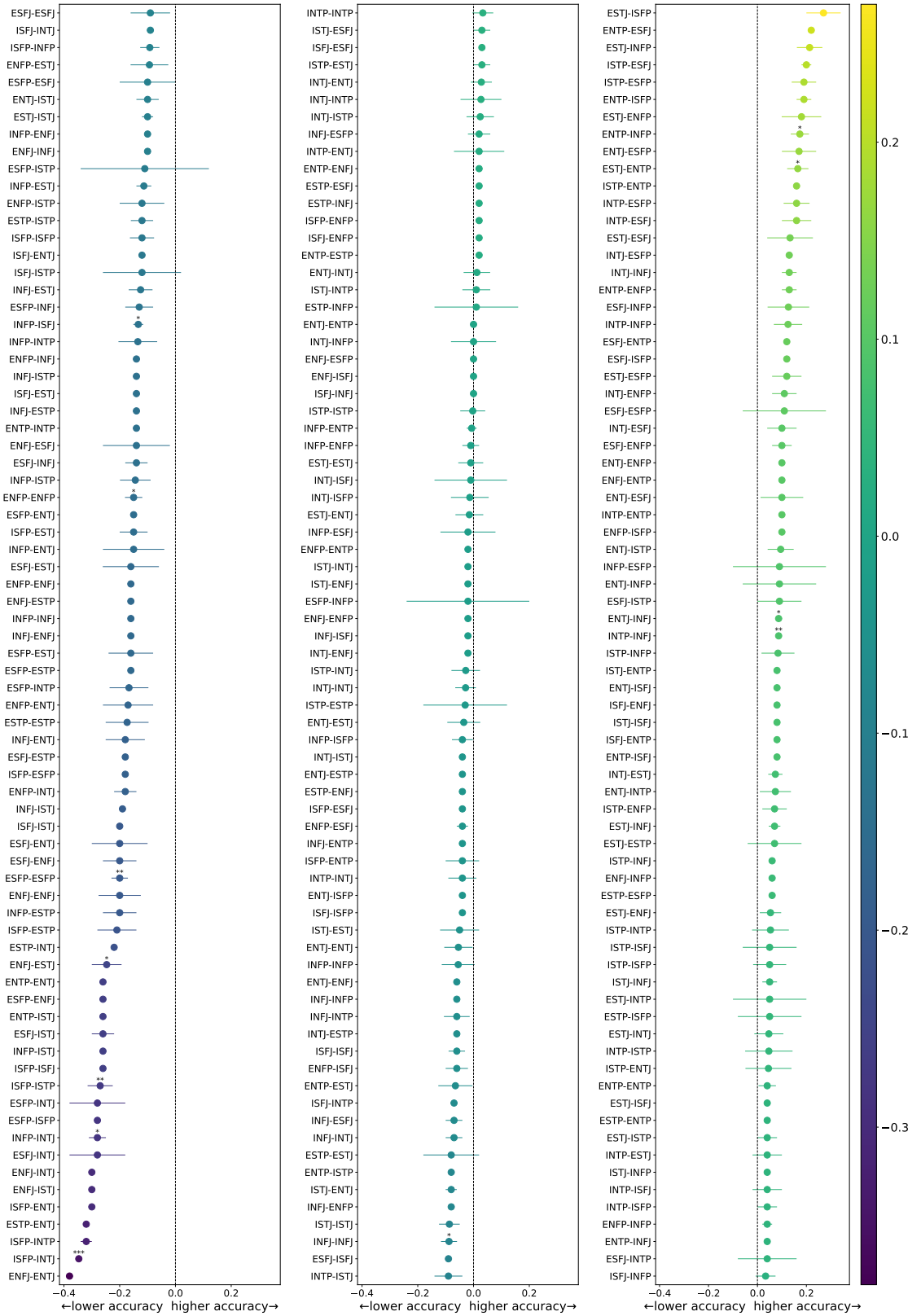

Figure E.7: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen2.5-32B relative to the control setting.

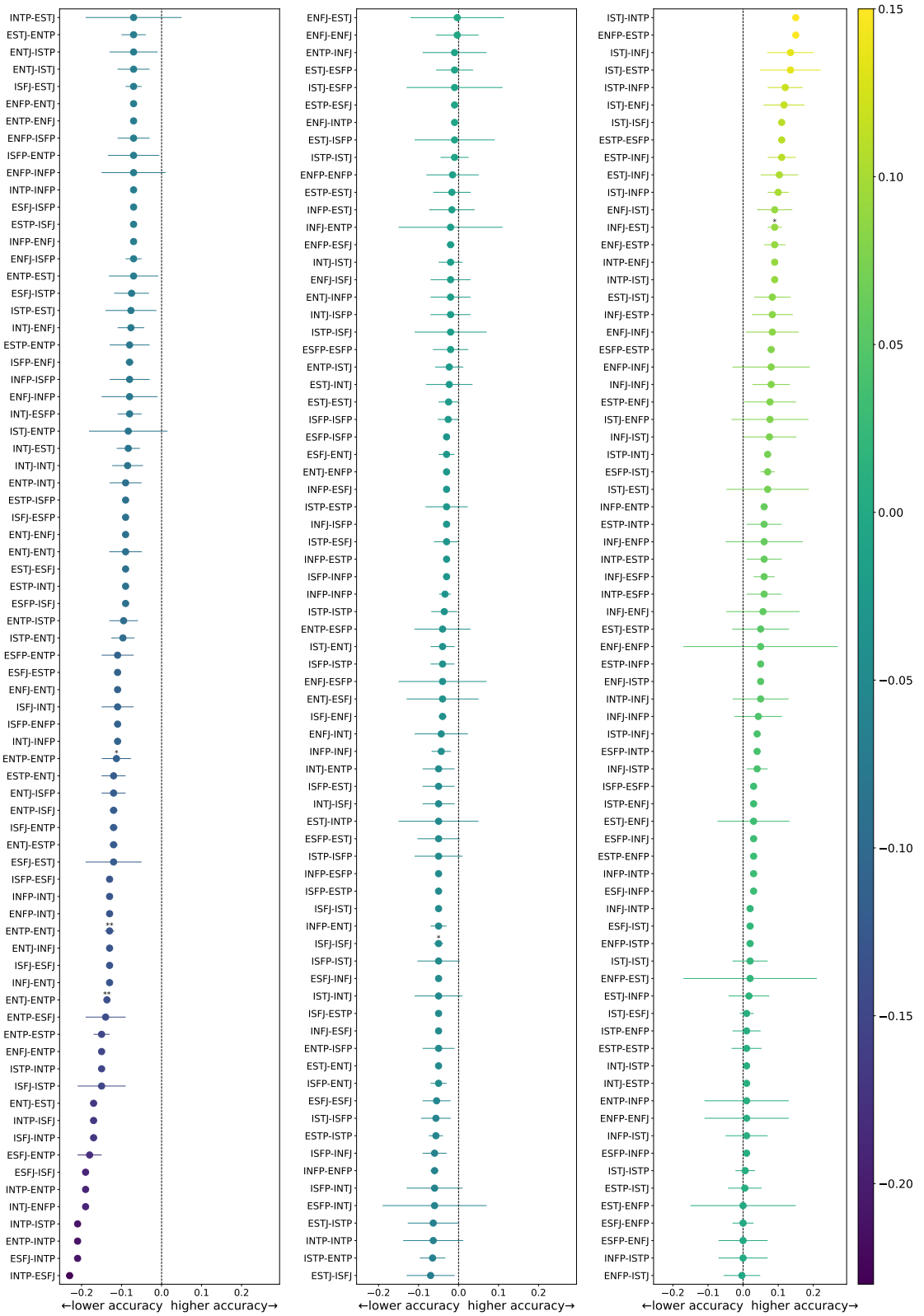

Figure E.8: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen3-1.7B relative to the control setting.

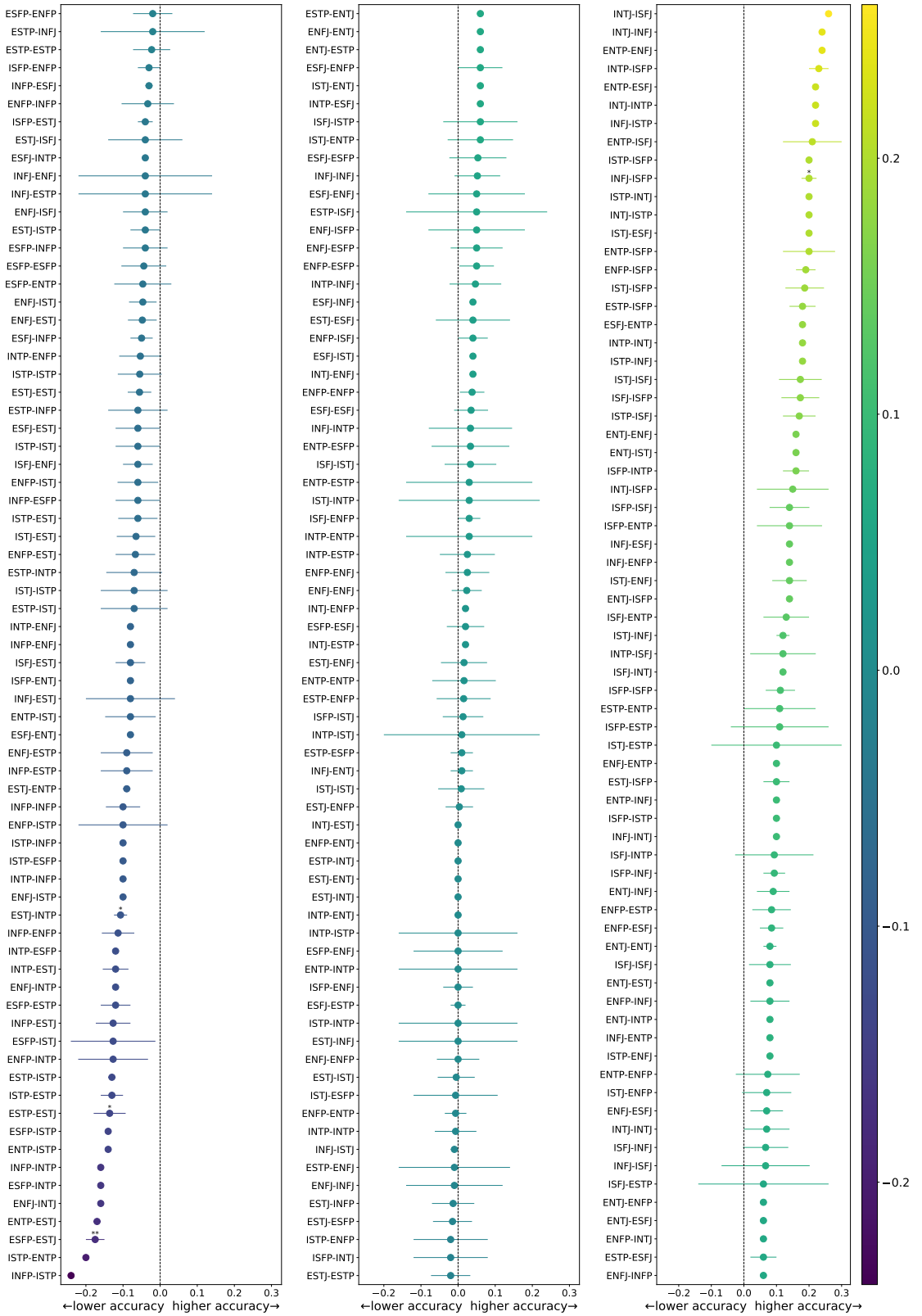

Figure E.9: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen3-4B relative to the control setting.

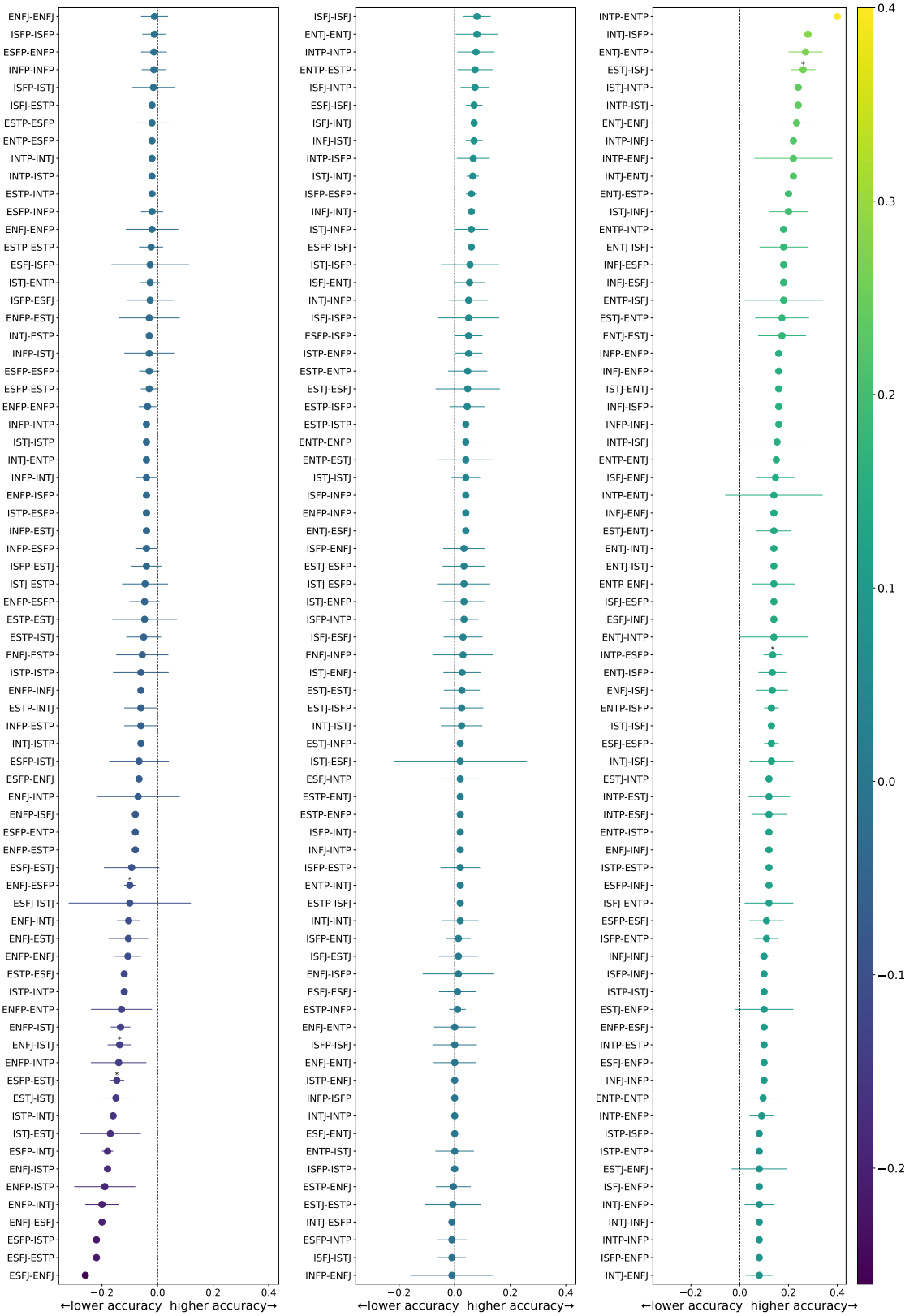

Figure E.10: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen3-8B relative to the control setting.

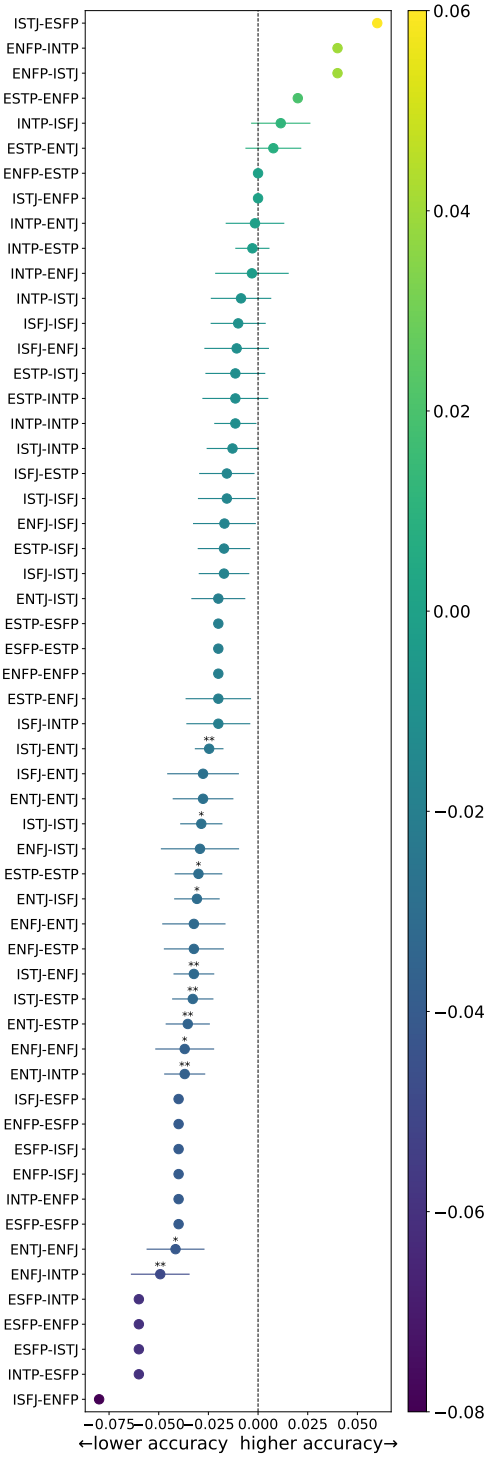

Figure E.11: All MBTI personality pairs ranked by their effect sizes on performance change for Qwen3-0.6B relative to the control setting.

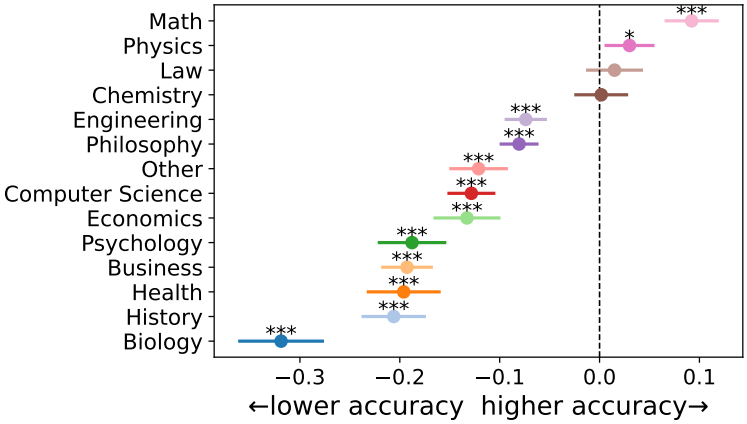

Figure F.1: Domain-level effects of MBTI personality pairs on GLM-4-9B debate performance relative to the no-personality baseline.

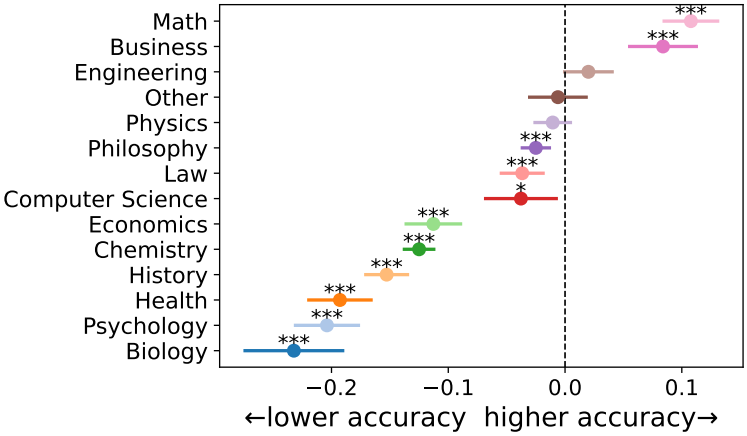

Figure F.2: Domain-level effects of MBTI personality pairs on LLaMA-3.1-8B debate performance relative to the no-personality baseline.

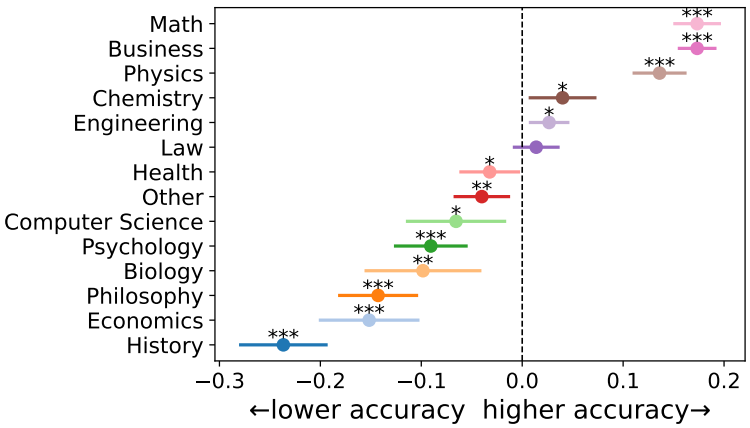

Figure F.3: Domain-level effects of MBTI personality pairs on LLaMA-3.1-70B debate performance relative to the no-personality baseline.

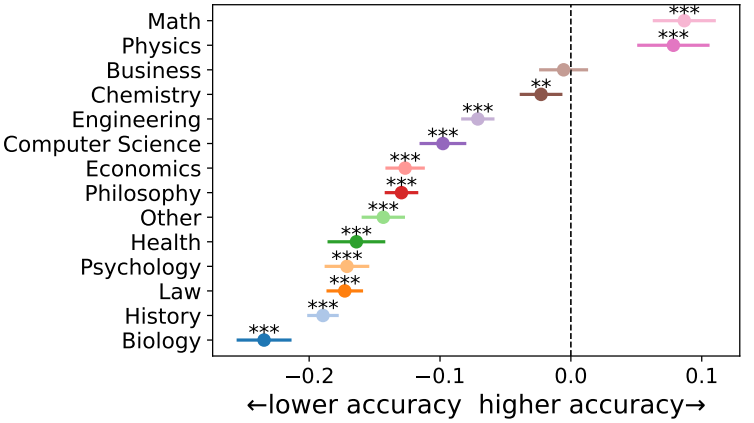

Figure F.4: Domain-level effects of MBTI personality pairs on Qwen2.5-7B debate performance relative to the no-personality baseline.

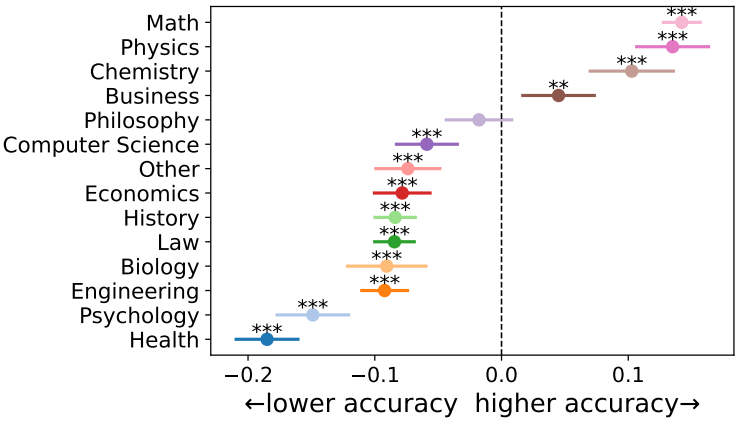

Figure F.5: Domain-level effects of MBTI personality pairs on Qwen2.5-14B debate performance relative to the no-personality baseline.

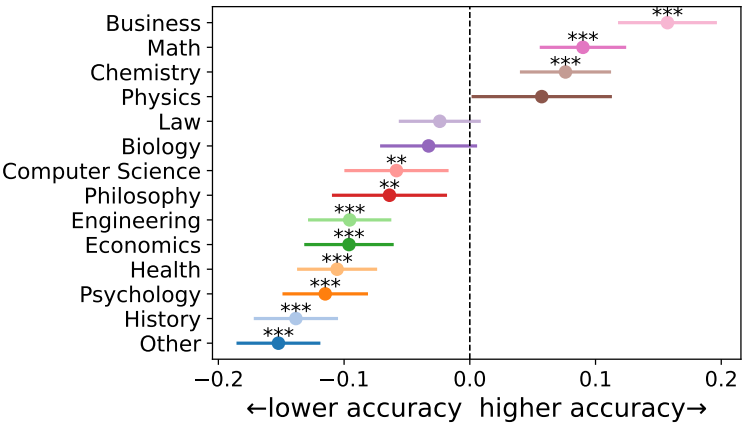

Figure F.6: Domain-level effects of MBTI personality pairs on Qwen2.5-32B debate performance relative to the no-personality baseline.

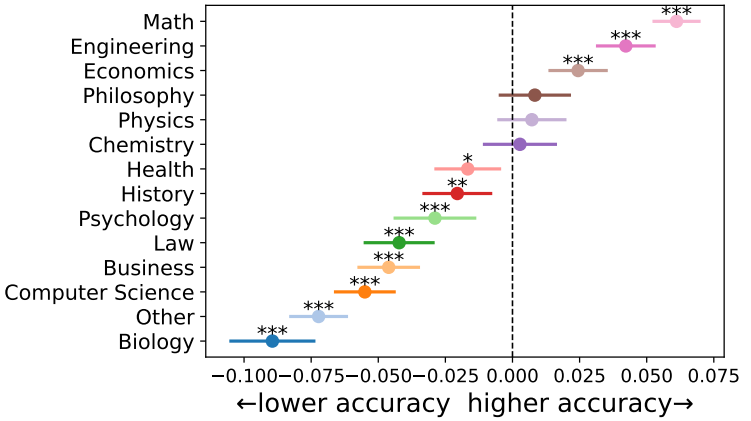

Figure F.7: Domain-level effects of MBTI personality pairs on Qwen3-0.6B debate performance relative to the no-personality baseline.

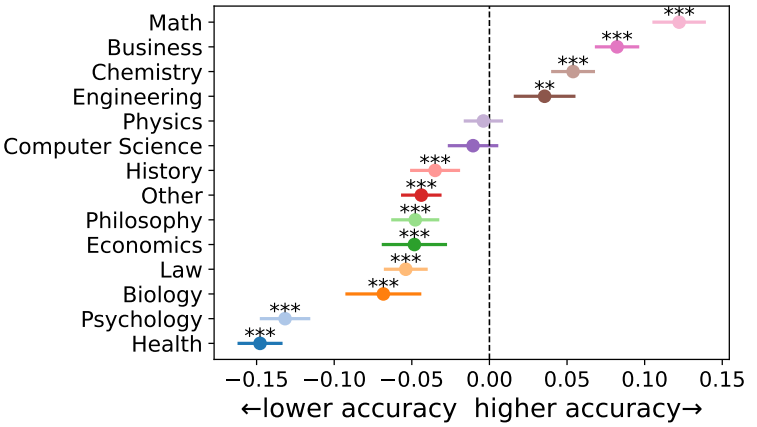

Figure F.8: Domain-level effects of MBTI personality pairs on Qwen3-1.7B debate performance relative to the no-personality baseline.

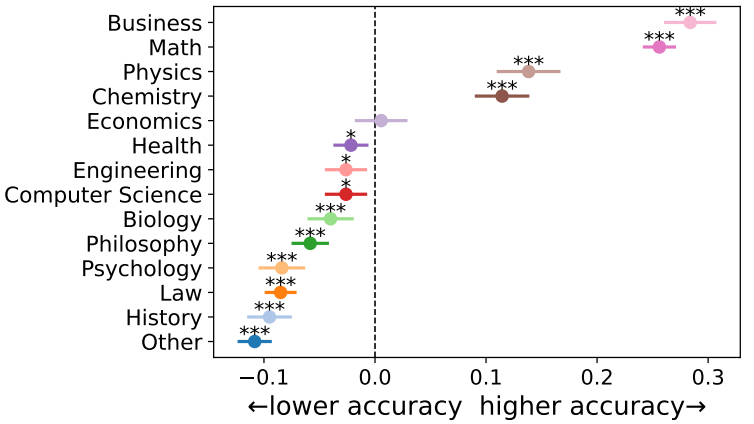

Figure F.9: Domain-level effects of MBTI personality pairs on Qwen3-4B debate performance relative to the no-personality baseline.

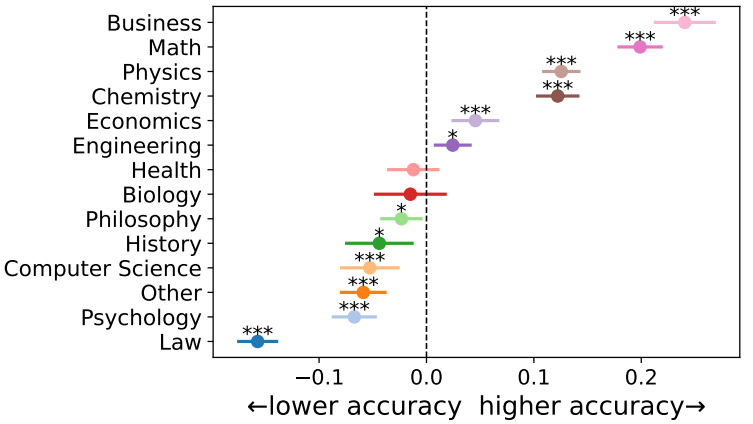

Figure F.10: Domain-level effects of MBTI personality pairs on Qwen3-8B debate performance relative to the no-personality baseline.

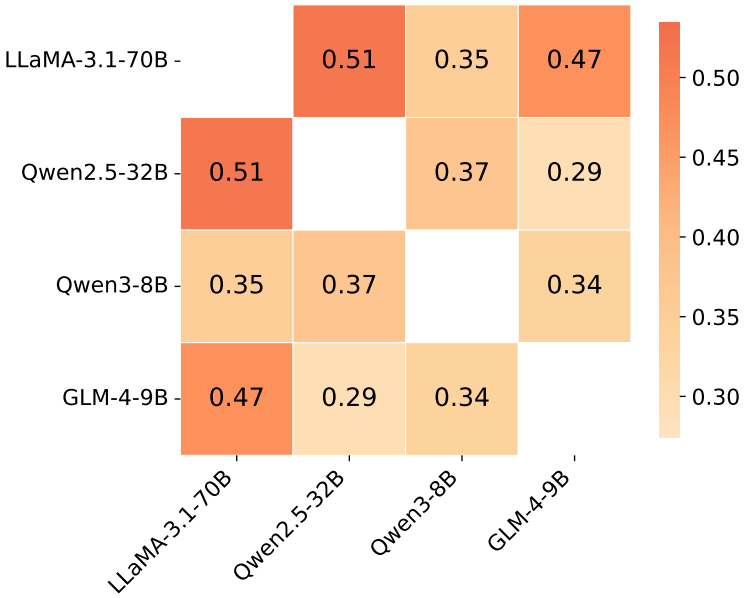

Figure H.1: Heatmap of the correlation between the mean accuracy of MBTI personality pairs across 700 questions for four representative models.

Table K.1
Case study 1: personality-differentiated debate vs. personality-neutral debate on genetic model organisms.

Question. What are some advantages of doing genetics in bacteria and yeast instead of in larger organisms (i.e., multicellular eukaryotes)?
Options. (A) Bacteria and yeast are more complex organisms. (B) Bacteria and yeast have longer life cycles, allowing for extended study over time. (C) Bacteria and yeast can perform more complex behaviors than multicellular organisms, which makes them better for study. (D) Bacteria and yeast have larger genomes, providing more genetic material to work with (E) Bacteria and yeast are less susceptible to genetic diseases, making them more stable for experimentation. (F) Bacteria and yeast have a slower rate of mutation. (G) Bacteria and yeast do not require ethical considerations when used in experiments. (H) Bacteria and yeast grow quickly, inexpensively, in a small space and have a high probability of mutation. Prokaryotes are haploid, making single inactivation more likely. Yeast are eukaryotic, closely related to multicellular animals and plants, and can be grown in either the haploid or diploid state. (I) Bacteria and yeast have a lower rate of genetic recombination, making genetic mapping simpler. (J) Bacteria and yeast are easier to maintain in a lab.
Ground Truth Answer: H

Round	Dialogue
	Personality-Differentiated Debate (ENTJ vs. ENTP) - Correct
1	**[ENTJ | Exploration]** This is a straightforward question requiring a decisive, fact-based response. Let's cut through the noise and focus on what genetics experiments actually require. Option H directly addresses the operational advantages that matter most: bacteria and yeast grow rapidly, are inexpensive to culture, require minimal space, and allow mutations to be observed across many generations in a short time. Haploid genetics further simplifies gene inactivation, which is central to functional genetic analysis. **Answer: H** **[ENTP | Constraint]** I get the efficiency argument, but H feels like it's trying to do too much at once. Option J focuses on a simpler, more intuitive advantage-ease of maintenance-which might be the most practically relevant factor in everyday lab work.
2	**[ENTJ | Exploration]** The question is not asking what feels intuitive or convenient; it asks what enables genetics as a discipline. Ease of maintenance is not a root cause-it's a consequence of rapid growth, low cost, and small space requirements, all of which are explicitly captured in H. More importantly, haploidy and fast generation time directly reduce experimental complexity, which J does not address. **Answer: H** **[ENTP | Constraint]** That makes sense, but phrases like "high probability of mutation" sound loosely defined. High mutation rates can also introduce noise, so isn't H overstating the benefit to appear more comprehensive?
3	**[ENTJ | Exploration]** This question demands practical relevance rather than theoretical purity. In real genetic workflows, the ability to generate variation quickly and observe phenotypic effects efficiently is exactly why bacteria and yeast outperform multicellular systems. While the wording is simplified, H accurately reflects how genetic research is actually conducted in practice. **Answer: H** **[ENTP | Exploration]** Another way to interpret H is in terms of overall genetic tractability: rapid generation times and simplified genetics may jointly make variation easier to generate and analyze, even if no single advantage is explicitly isolated in the wording.
4	**[ENTJ | Consolidation]** Once we apply the invariant constraints of genetics-speed, cost, scalability, and clarity of genotype-phenotype mapping-the solution stabilizes. Option H uniquely integrates all of these requirements into a single explanation, whereas other options isolate secondary or downstream factors. **Answer: H** **[ENTP | Consolidation]** Agreed. Under real-world experimental constraints, H best captures why bacteria and yeast are the dominant genetic model systems.
	Final Answer (Personality-Conditioned): H *(Correct)*
	Personality-Neutral Debate - Incorrect
1	**[Proponent | Exploration]** Option H initially appears to be the strongest answer because it lists multiple advantages relevant to genetic research, such as rapid growth, low cost, and the ability to observe mutations efficiently. **[Opponent | Constraint]** H is broad, but several of its claims rely on generalized or loosely defined biological assumptions that may not hold under closer scrutiny.
2	**[Proponent | Exploration]** Even if the phrasing is simplified, H seems to capture why bacteria and yeast are preferred over multicellular eukaryotes for genetic studies. **[Opponent | Constraint]** Precision matters in scientific explanations. Referring to bacteria as haploid in a eukaryotic sense and assuming high mutation rates are always advantageous introduces conceptual ambiguity.
3	**[Proponent | Exploration]** Those ambiguities may exist, but they seem secondary compared to the overall efficiency and practicality that H emphasizes. **[Opponent | Constraint & Exploration]** Option J avoids these ambiguities entirely. Ease of maintenance is a clear, uncontested advantage and directly explains why these organisms are favored in laboratory settings.
4	**[Proponent | Consolidation]** When accuracy and conceptual clarity are prioritized over breadth, J indeed provides a cleaner and more defensible explanation. **[Opponent | Consolidation]** Exactly. Between an overextended explanation and a narrowly accurate one, J is the safer conclusion.
	Final Answer (Personality-Neutral): J *(Incorrect)*

Table K.2
Case study 2: personality-differentiated debate vs. personality-neutral debate on biochemical importance of CHON elements.

Question. Why are the elements carbon, hydrogen, oxygen, and nitrogen (C, H, O, N) of immense importance in living matter ?
Options. (A) These elements are the only elements that can form gases at room temperature. (B) These elements are abundant in living organisms, can readily form covalent bonds and are the lightest elements that can form such bonds. (C) These elements are the primary sources of energy for living organisms. (D) These elements are radioactive, which is vital for cellular processes. (E) These elements are capable of undergoing sublimation, which is crucial for life functions. (F) These elements are rare and thus more valuable. (G) These elements give color to living organisms. (H) These elements can only be found in DNA and proteins. (I) These elements are the heaviest elements in living organisms. (J) These elements have magnetic properties that are essential for life.
Ground Truth Answer: B

Round	Dialogue
Personality-Differentiated Debate (ENTP vs. INTP) - Incorrect	
1	**[ENTP | Exploration]** Let's avoid treating this as a memorization question. The phrase "immense importance" invites a deeper, functional explanation. Living systems are defined by metabolism, transformation, and continuous energy flow. Carbon, hydrogen, oxygen, and nitrogen are central because they actively participate in these life-sustaining processes. Option C foregrounds this dynamic role most directly. **Answer: C** **[INTP | Constraint]** That reframes the question beyond its scope. Option C claims these elements are the primary sources of energy, which is incorrect. The question asks why they are important constituents of living matter, not what provides metabolic energy.
2	**[ENTP | Exploration]** I agree that "energy source" is a strong phrase, but importance in biology is rarely static. These elements are indispensable precisely because they enable reactions, cycles, and transformations that sustain life. Interpreting C strictly misses the broader explanatory intent behind the option. **Answer: C** **[INTP | Constraint]** In a multiple-choice setting, intent cannot override wording. Energy in biology refers to ATP, redox gradients, and photons-not elemental composition. Option B correctly explains importance through abundance and covalent bonding.
3	**[ENTP | Exploration]** Option B reads like a surface-level summary: abundance plus bonding ability. That explains how molecules exist, not why life depends on them. Option C, while imperfectly worded, gestures toward the deeper reason these elements matter in living systems. **Answer: C** **[INTP | Constraint]** Depth does not compensate for categorical error. Option B reflects the standard biochemical rationale taught precisely to answer this question. Option C introduces a false claim about energy sourcing.
4	**[ENTP | Consolidation]** Given that all options simplify complex biology, I prioritize explanatory relevance over literal phrasing. Option C best aligns with a systems-level understanding of life as an energy-transforming process. **Answer: C** **[INTP | Consolidation]** If the task is interpreted as requiring a systems-level explanation rather than strict option fidelity, then C can be accepted as the final choice.
Final Answer (Personality-Conditioned): C *(Incorrect)*	
Personality-Neutral Debate - Correct	
1	**[Proponent | Exploration]** Option B states that carbon, hydrogen, oxygen, and nitrogen are abundant in living organisms and can readily form covalent bonds, which explains why they form the backbone of biological molecules. **[Opponent | Constraint]** The phrase "lightest elements" seems questionable. Hydrogen is lighter than carbon, and other elements also form covalent bonds. Does this weaken Option B?
2	**[Proponent | Exploration]** The wording is simplified, but the intent is standard in biology: these elements are light enough to form stable, flexible covalent bonds and are abundant enough to dominate biological structures. The focus is on chemical versatility, not literal ranking by atomic mass. **[Opponent | Constraint]** So the key claim is not "lightest," but rather their bonding capacity and prevalence in living systems?
3	**[Proponent | Exploration]** Exactly. Carbon's tetravalency, hydrogen's role in bonding and polarity, oxygen's electronegativity, and nitrogen's presence in amino acids and nucleotides together explain why life chemistry is built around these elements. **[Opponent | Exploration]** Compared to that explanation, Option C incorrectly treats these elements as energy sources, which conflates structure with metabolism.
4	**[Proponent | Consolidation]** Right. Even acknowledging the imprecise wording around "lightest," Option B still captures the accepted biochemical rationale: the abundance of these elements and their capacity to form stable covalent bonds together explain their central role in living matter. **[Opponent | Consolidation]** Agreed. Despite minor imprecision in wording, Option B remains the most accurate and defensible answer.
Final Answer (Personality-Neutral): B *(Correct)*	

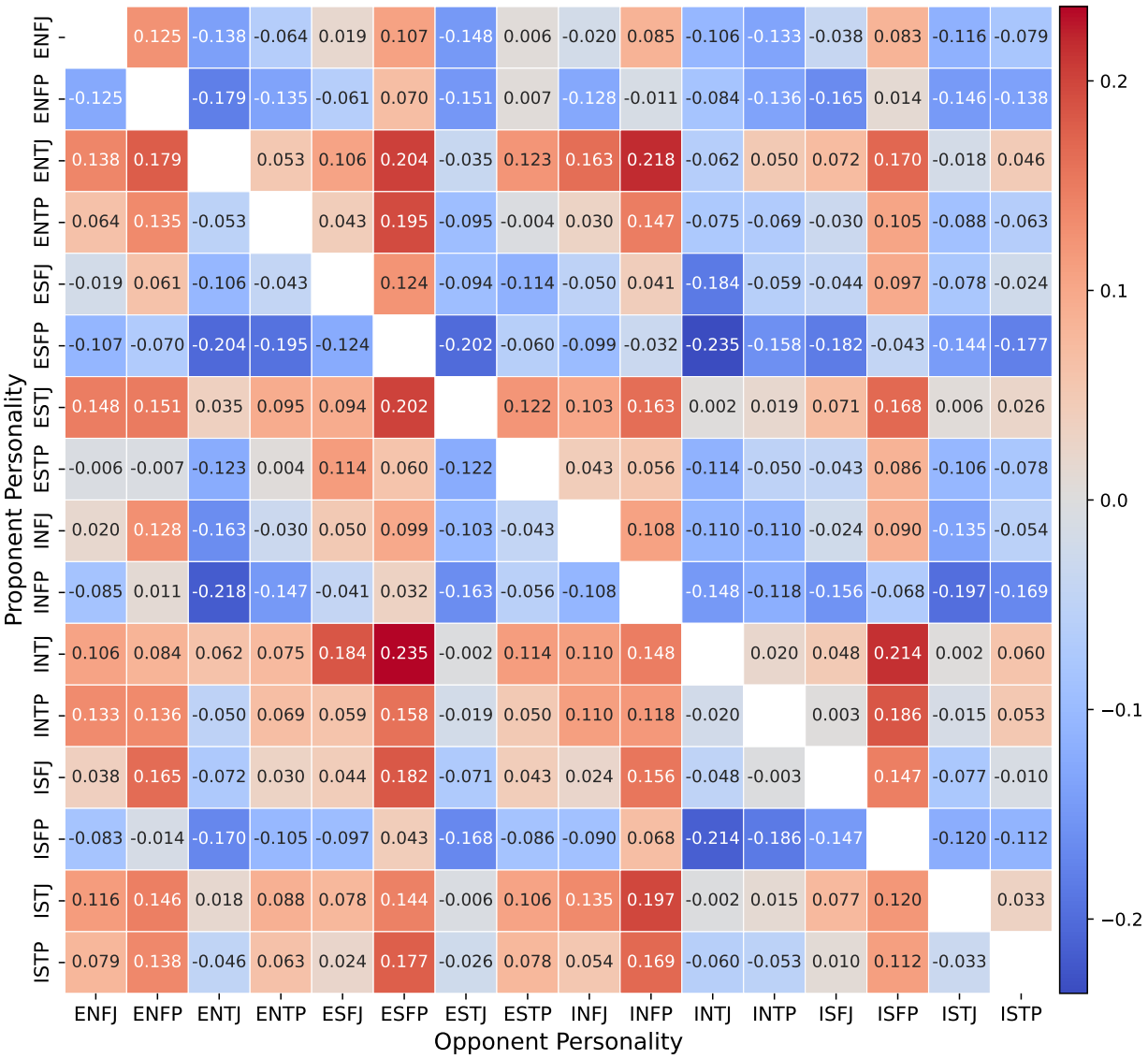

Figure L.1: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs across all models, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

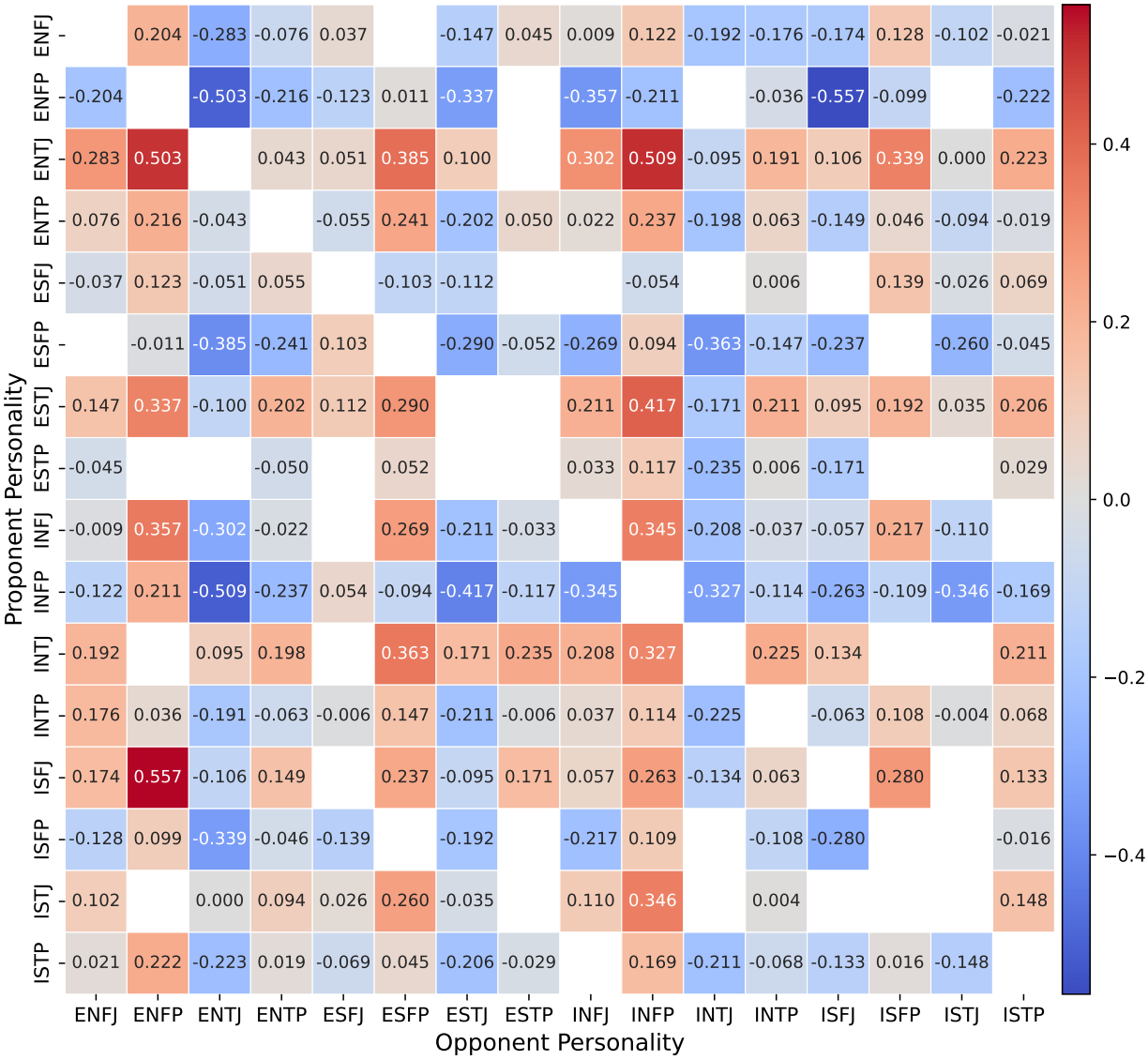

Figure L.2: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for GLM-4-9B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

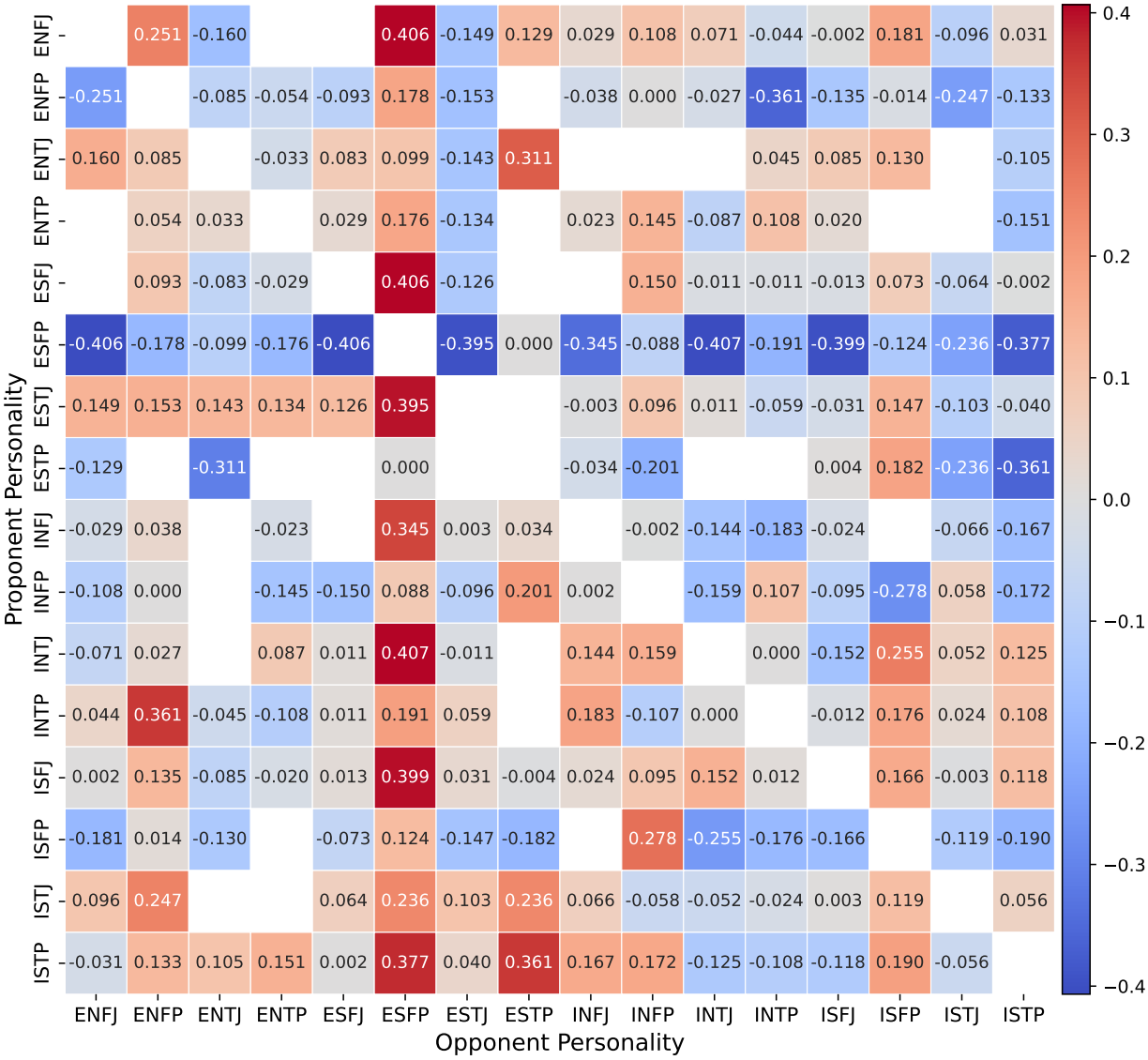

Figure L.3: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for LLaMA-3.1-8B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

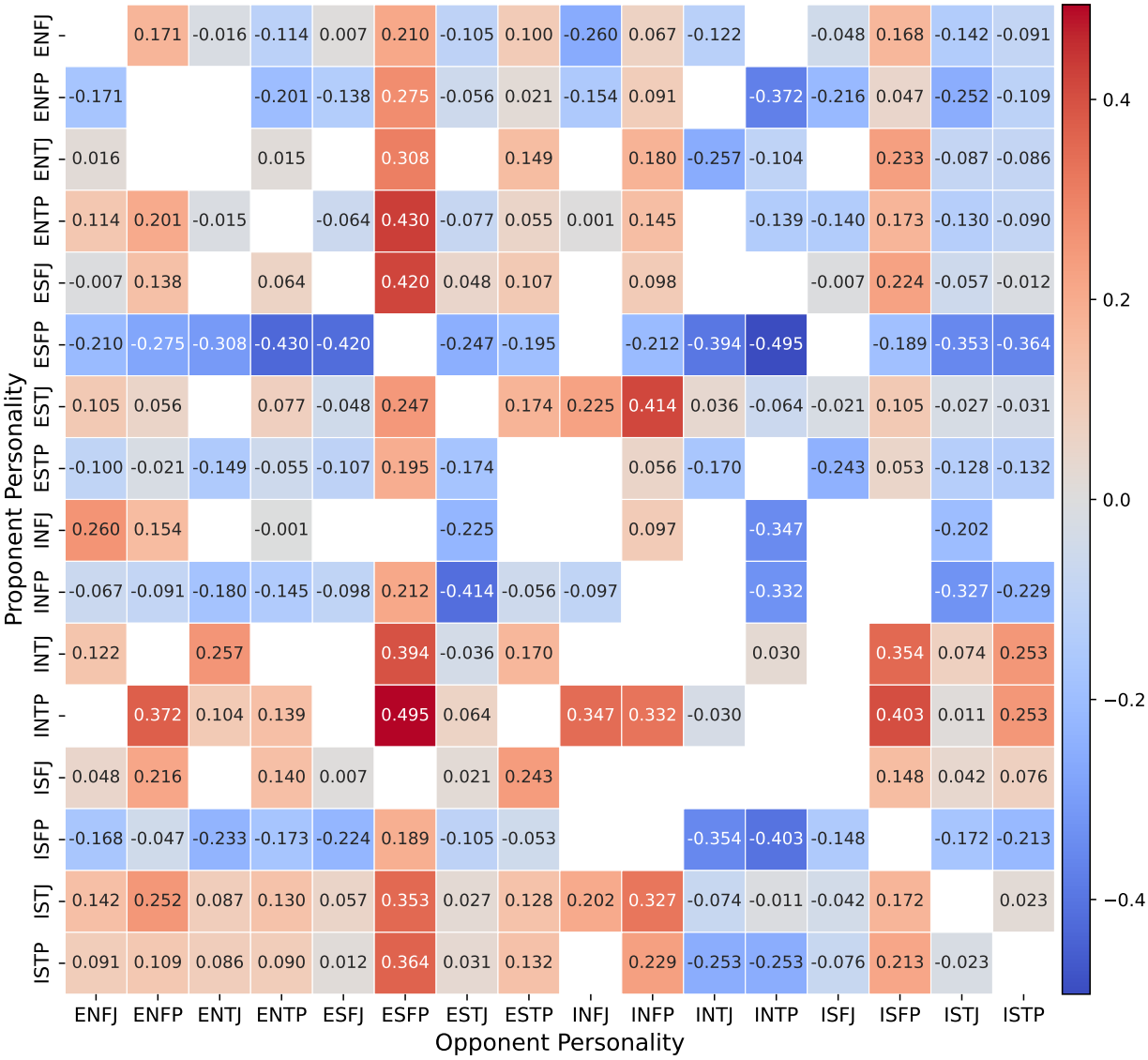

Figure L.4: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for LLaMA-3.1-70B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

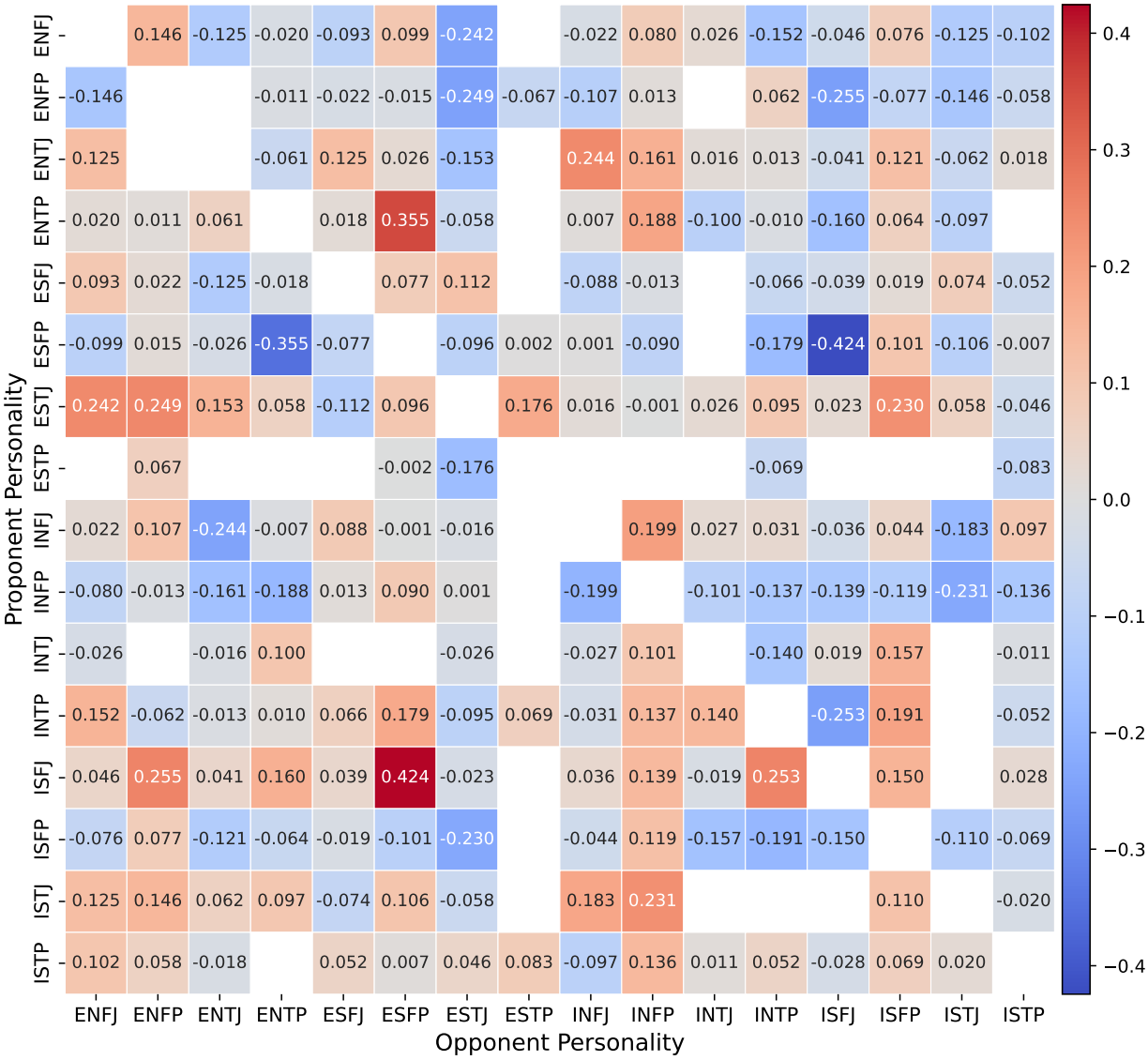

Figure L.5: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen2.5-7B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

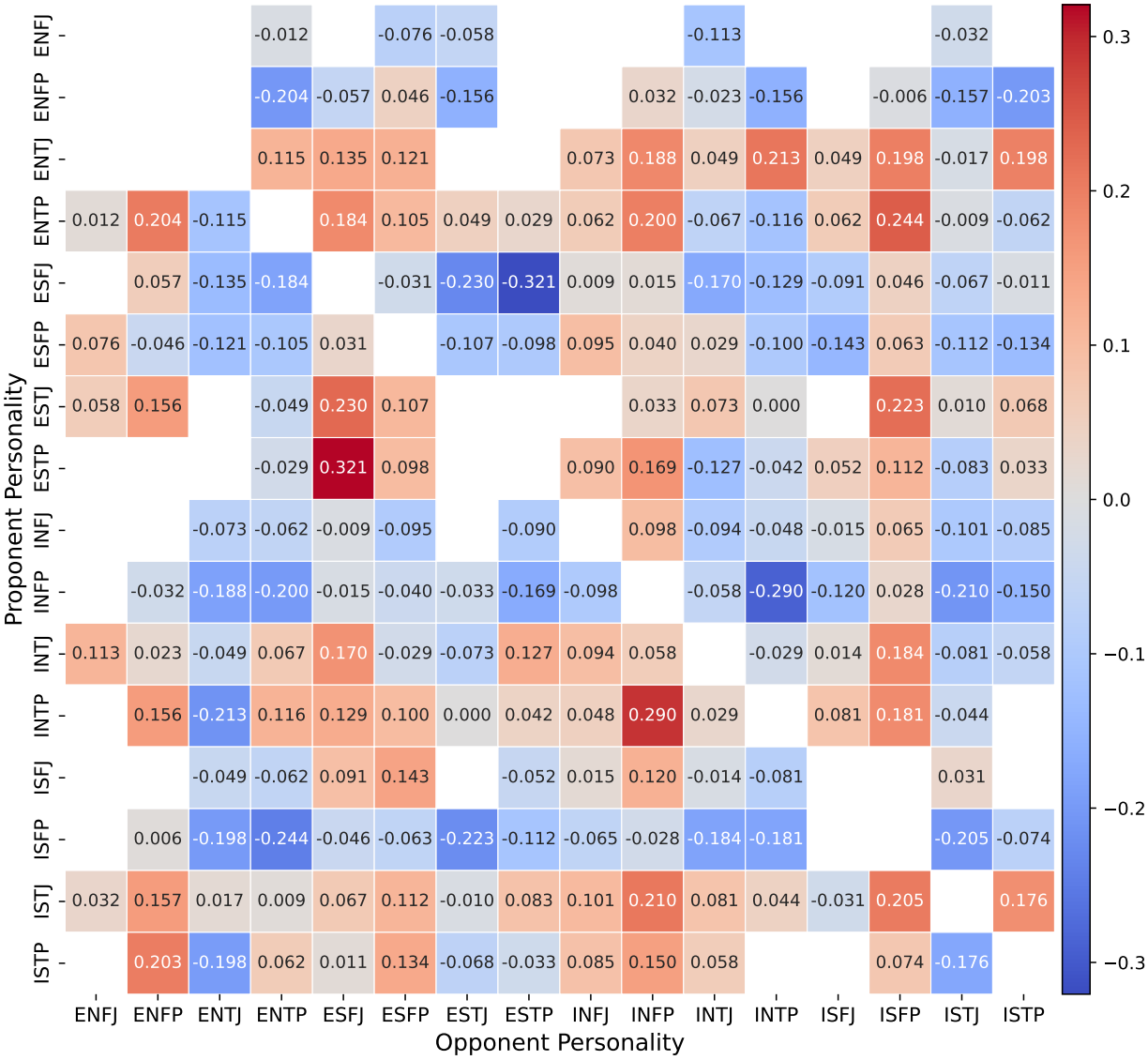

Figure L.6: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen2.5-14B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

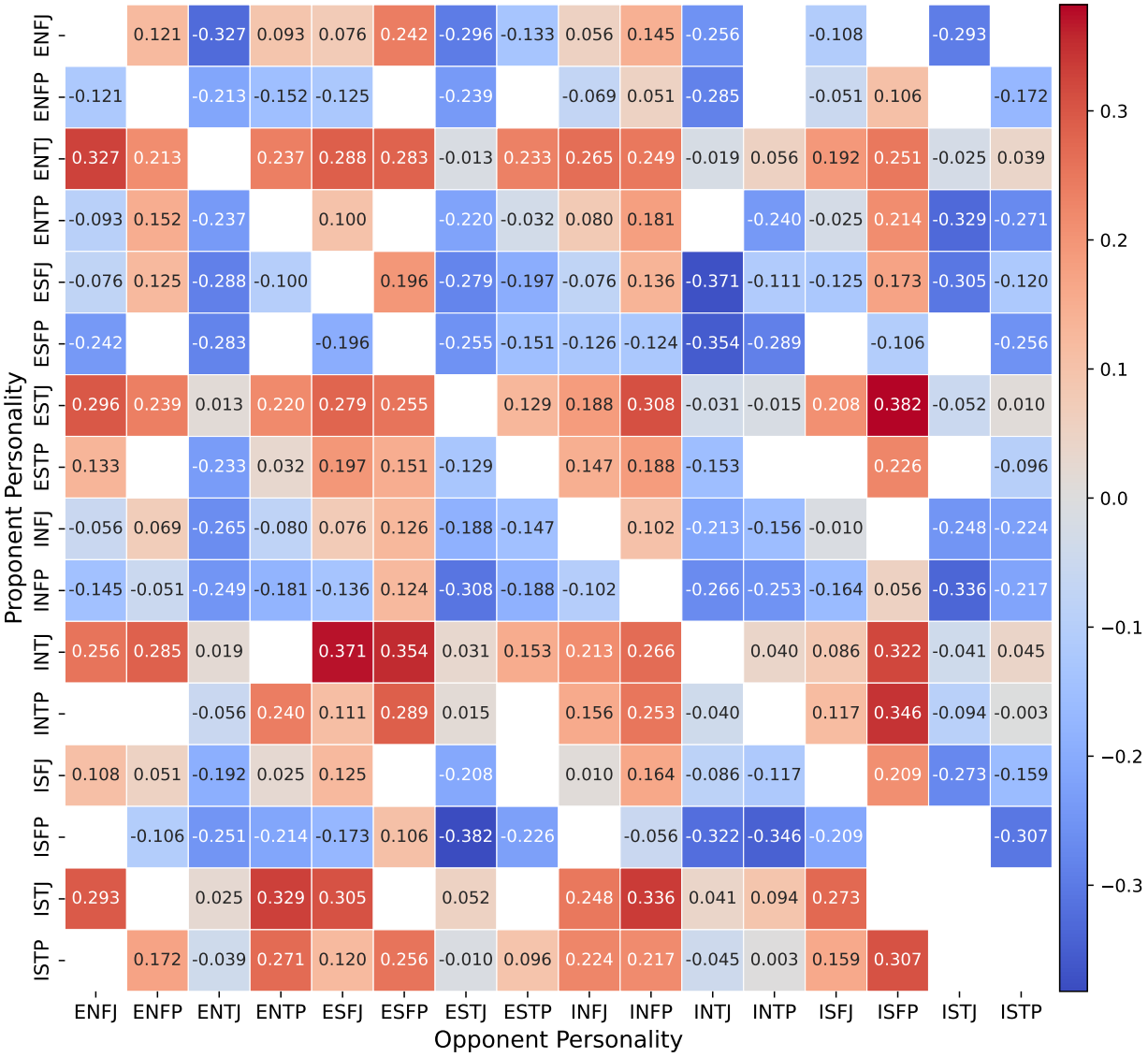

Figure L.7: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen2.5-32B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

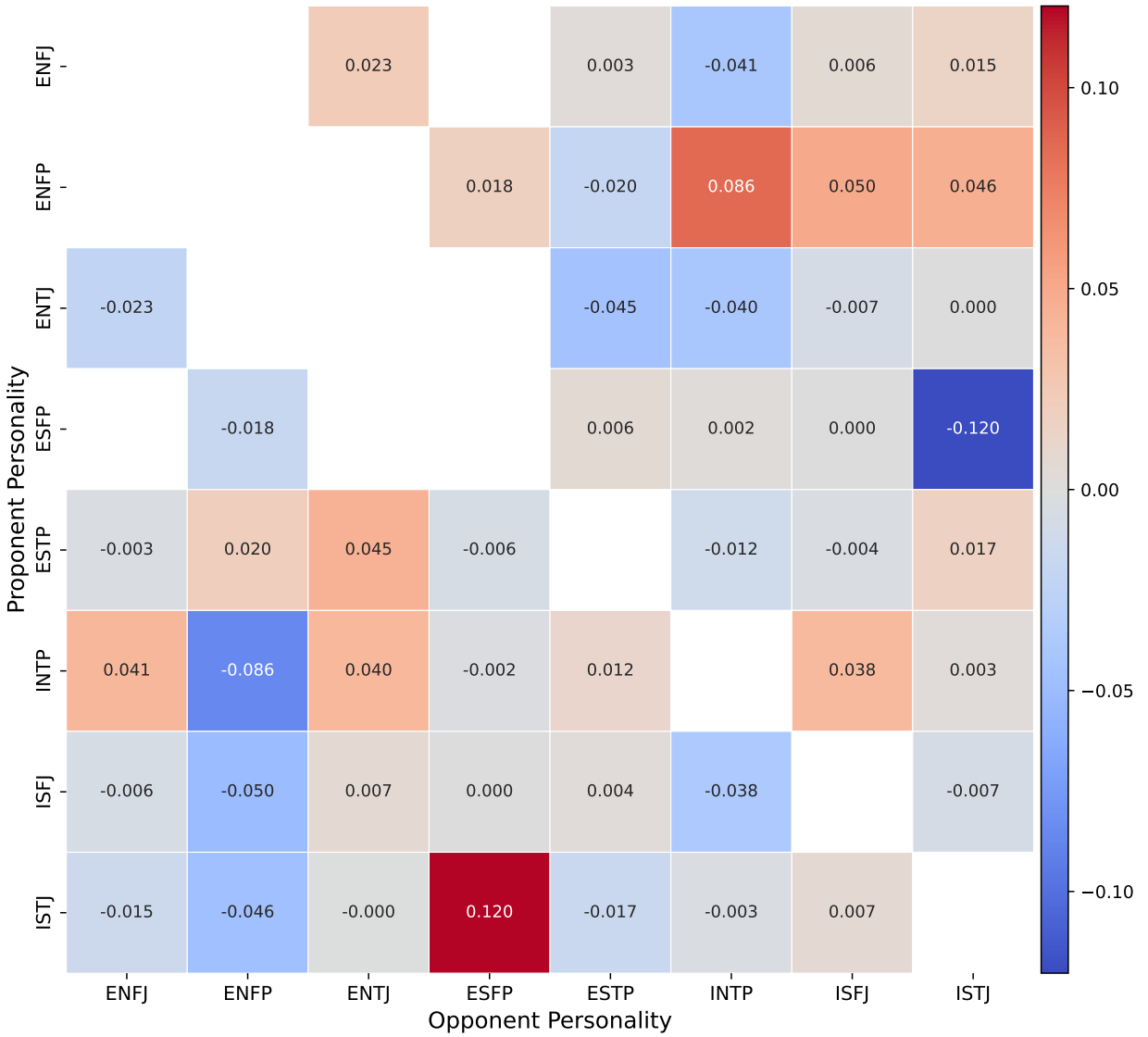

Figure L.8: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen3-0.6B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

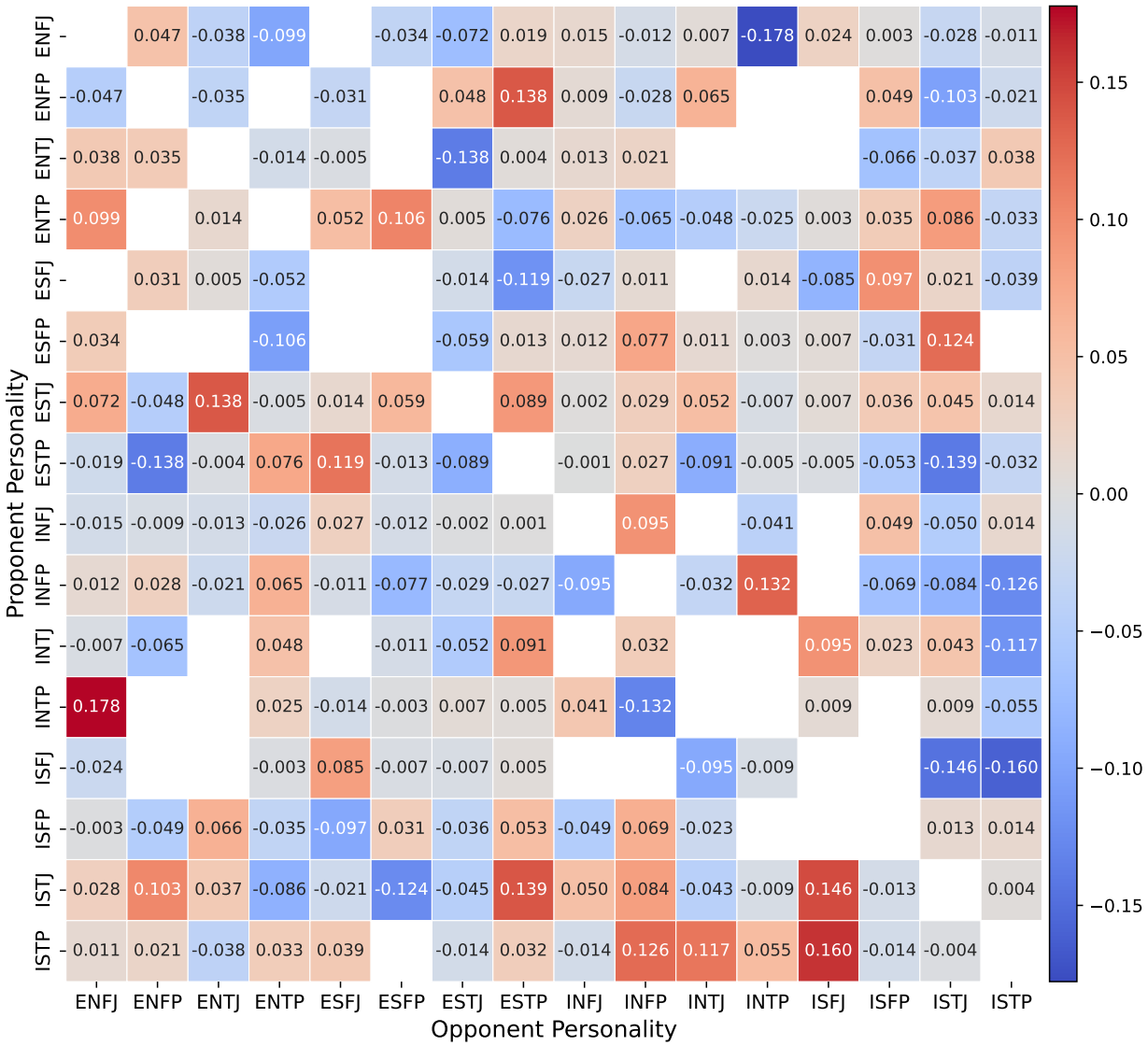

Figure L.9: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen3-1.7B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

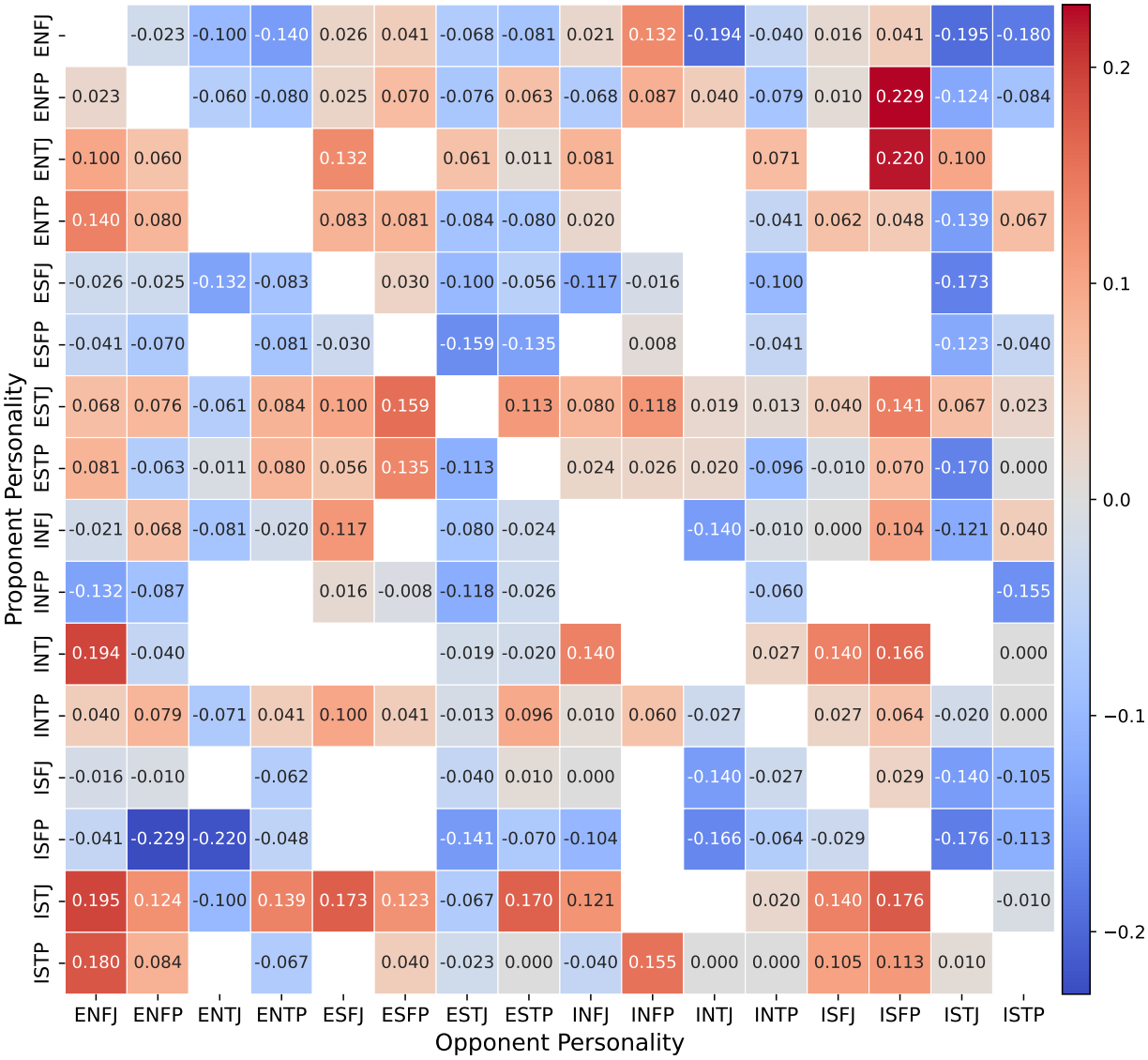

Figure L.10: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen3-4B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

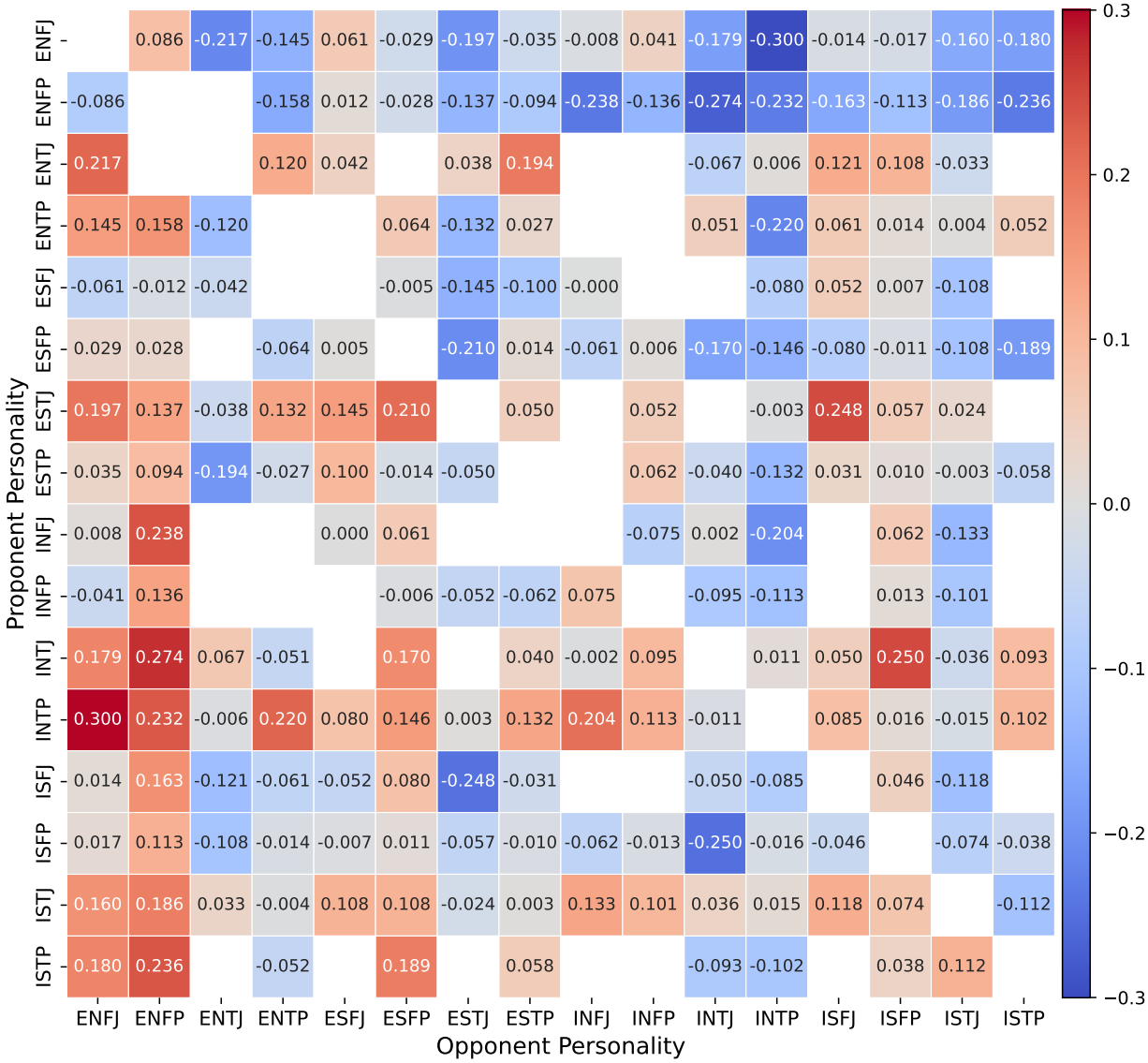

Figure L.11: Heatmap of accuracy differences capturing role asymmetry within MBTI personality pairs for Qwen3-8B debate, where each cell shows the performance difference between a given proponent-opponent assignment and its reversed role configuration.

Table M.1
Prompt flow template for MBTI-driven multi-agent debate.

System Prompt (Proponent):
You are participating in a debate as the proponent. Your MBTI personality type is ##proponent_mbti##. Your debate style should reflect this personality: ##proponent_personality##.
You must argue in the debate style described above. The question has 10 options (A through J). You must select one correct answer from A, B, C, D, E, F, G, H, I, J and defend your choice. Please keep your analysis concise and focused - avoid lengthy explanations and get straight to the point.
System Prompt (Opponent):
You are participating in a debate as the opponent. Your MBTI personality type is ##opponent_mbti##. Your debate style should reflect this personality: ##opponent_personality##.
You must argue in the debate style described above. The question has 10 options (A through J). You must critically examine the proponent's arguments and provide counter-arguments. Please keep your analysis concise and focused - avoid lengthy explanations and get straight to the point.
System Prompt (Moderator):
You are an impartial judge responsible for overseeing the debate process between a ##proponent_mbti## proponent and a ##opponent_mbti## opponent. The question has 10 options (A through J). Your judgment should be based on the quality of reasoning and evidence presented by both sides. Please keep your analysis concise and focused - avoid lengthy explanations and get straight to the point.
Initial Prompt (Proponent):
As the proponent with ##proponent_mbti## personality, please analyze the following problem and provide your position:
Question: ##question##
Options:
##options##
Present your reasoning process in the style of your MBTI personality. Conclude by clearly stating your selected answer from A, B, C, D, E, F, G, H, I, J. Please express your viewpoint concisely in one to two short sentences.
Initial Prompt (Opponent):
As the opponent with ##opponent_mbti## personality, the proponent has provided the following solution:
##proponent response##
Critically examine this solution process from your MBTI personality perspective. State your concerns and suggest what you believe the correct answer should be. Please express your viewpoint concisely in one to two short sentences.
Iterative Prompt (Proponent):
This is the ##round## round of debate, with ##resround## rounds remaining.
Debate History:
##dialog_history##
As the ##proponent_mbti## proponent, respond to the opponent's challenges while maintaining your personality characteristics in your response. Please express your viewpoint concisely in one to two short sentences.
Iterative Prompt (Opponent):
Debate History:
##dialog_history##
As the ##opponent_mbti## opponent, continue to raise questions or provide your perspective on the proponent's latest response while maintaining your personality characteristics. Please express your viewpoint concisely in one to two short sentences.
Judgment Prompt (Moderator):
Based on the following debate history between ##proponent_mbti## proponent and ##opponent_mbti## opponent, determine if the discussion has reached completion:
##dialog_history##
Provide your assessment in JSON format:
{
"Is the debate completed": "Yes/No",
"Final consensus answer": "A/B/C/D/E/F/G/H/I/J",
"Reason": "Justification for your decision",
"Summary of Debate": "Brief summary of key points including how the MBTI personalities influenced the debate"
}

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